

Phonetics and Phonotactics

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1. Transcription conventions

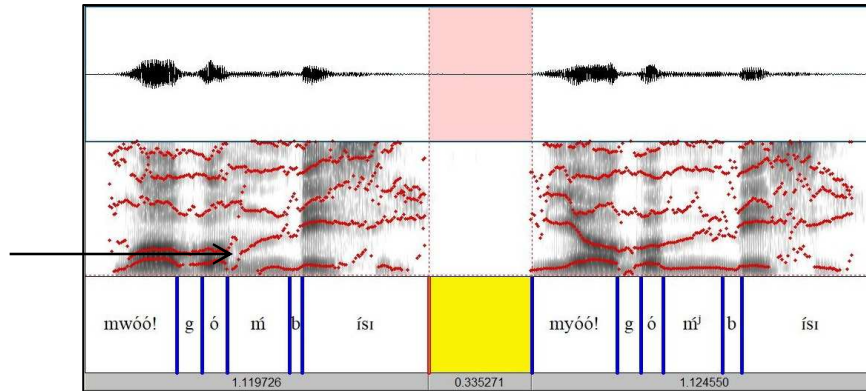
Data in this work are presented in a compromise between IPA phonemic transcriptions and (a kind of) augmented spelling. That implies two specific things. First, less or more information about pronunciation may be included in a given example, depending on what level of information is being conveyed. The implied transcription is not designed to be strictly phonemic or strictly phonetic. Second, ordinary Swahili-style spelling is pursued as much as is practical, thus you will see *ch*, *y* and not *tʃ*, *j* – you will see *j* but only for the voiced palatal affricate, so you won't see *dʒ*. There are significant sounds in Logoori which are outside the grasp of Swahili-spelling such as tones or the vowels [ɪ, ʊ], so those are also part of the necessary transcriptional system. At <https://Languagedescriptions.github.io/Logoori/WritingtheLogoorilanguage> there is a lay explanation regarding the need for special symbols such as ɪ, ʊ, ɲ along with contrastive tone marks (á, í vs a, i and the use of '). That document includes some sound files, designed to familiarize a speaker of the language having no training in linguistics with an indication of what these symbols represent. As well, the linguist reader can hear what some samples of the language sound like. Other soundfiles can be found in the online dictionary at <https://languagedescriptions.github.io/Logoori/OnlineDictionary/>. Because the goal of the present work is to scientifically record as much information about the language as is practical, including details of pronunciation, a given word may be written in many different ways in this work, because there are many different actual pronunciations of words that are beyond the reach of spelling.

In terms of segment values, the following translations between the present work and IPA are to be applied.

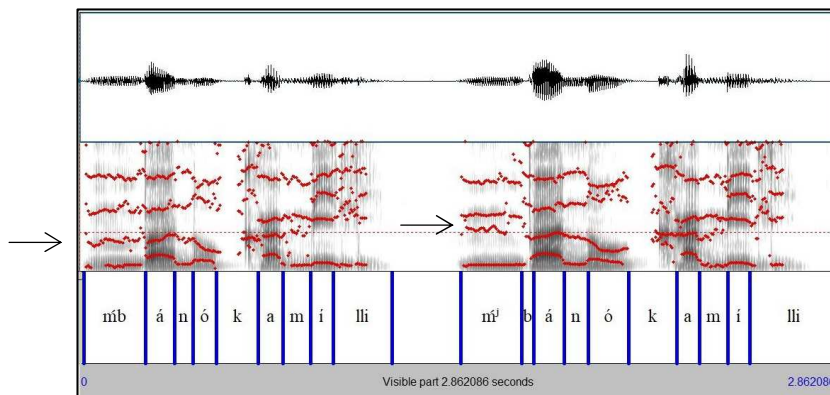
<i>This work</i>	<i>IPA</i>
ch	tʃ
j	dʒ
sh	ʃ
y	j
ny	nj ~ nʲ, ɲ
ɲ	variable nj, ɲ, ɲj

Long vowels are notated by doubling the vowel letter (*koroota* ‘to dream’), likewise geminate consonants are indicated by doubling the first letter (*ɪddɪjɪ* ‘window’). There are no underlying geminates in the language, but many speakers employ derived geminates arising from reduction of *ri+C*, *ro+C*, for example [ɪddɪku] ‘day’ (EM,RK,BK,FI), alternating with (*ɪ*)*rídiku* for these speakers; *inneke* ~ *rineke* ‘*Syzygium cordatum*’ (BK); *ittigmyu* ~ (*ɪ*)*ritigmyu* ‘heel’ (EM,BK). All speakers appears to at least potentially have geminate *l* from /rVr/ (if indeed the phoneme is /r/ rather than /l/), e.g. *lláánde* ‘*Stephania abyssinica*’ (/ri-ráánde/, cf *maráánde* ‘S.A. plants’), *llími* ‘tongue’ (/ro-rími/). Geminate [vv] also arises (again, for at least some speakers) by reduction of /vi+v/ (vvírí ‘2 (cl. 8)’ /vi-vírí/) and /vo+v/, vvírí ‘2 (cl. 14)’ /vo-vírí/). Some speakers have geminate consonants in ideophones.

Non-contrastive details are included at various points in the work, providing information about the pronunciation of particular tokens (or dialects). For example, a superscripted pre-consonantal glide e.g. [ʷmbí] or [ɪmbí] ‘bad (cl. 4)’, [ʷmbí] ‘bad (cl. 3)’, is encountered in some examples from speaker BK – this will surprise linguists since such palatalization is not a typical feature of Bantu. This notation indicates a palatal offglide (versus labial offglide or, possibly, no offglide) from the preceding vowel. This feature was noticed and recorded before an analysis of the phenomenon was available. This palatalization seems to always be the trace of a deleted front prefix vowel /i, ɪ/. An example is [myóó¹gó m^ybíɪ] ‘raw cassavas’, underlyingly /mi-ógo ɪ-mi-víɪ/, therefore the surface form might reflect a non-phonological merger of the vowel sequence /o#ɪ/. Or it may be a trace of the vowel /i/ in the class prefix /mi/. The first analysis was not initially considered, since it seemed that BK lacked the augment, but subsequent analysis revealed that the augment may be underlyingly present yet deleted (almost everywhere, for this speaker). The same phenomenon has been observed with a few other (‘remote interaction’) speakers. The effect can be seen in the following spectrograms. The first example contrasts *mwóó¹gó mbíɪ* ‘raw cassava’ versus *myóó¹gó m^ybíɪ* ‘raw cassavas’ from speaker FA. In the former utterance (with non-palatalized *m*), the second formant during *m* is initially quite low and rises significantly within the nasal, and in the second utterance F₂ starts higher and rises less.



In the following example from FA, *mbánó kamilli* ‘proper knife’ versus *m^ybánó kamilli* ‘proper knives’ (or, *^ymbánó kamilli*), the utterance-initial difference [m] versus [m^y] again shows that palatalized [m^y] has a higher F₂, averaging 1590 Hz versus 1418 Hz in the case of [m].



This effect has not been observed with other speakers, which may reflect a difference in the underlying distribution of the augment across speakers (some speakers have analyzed the augment out of existence). In the case of EM, augment deletion is dispreferred, and his strategy for phrasal hiatus reduction favors retention of the second vowel (the augment). For EM, we have *mwóó! g-óm' bísɪ* ‘raw cassava’ versus *myóó! g-ím' bísɪ* ‘raw cassavas’ where there is a very obvious difference in surface vowels of the adjective. At any rate, this was a minor trend in the data, attested rarely enough that a systematic study is not currently possible.

Non-identical vowel sequences poses a minor transcriptional problem. Such sequences arise in three contexts. One is as the realization of /Vyɪ/, where y may be phonetically elided before [ɪ], giving rise to forms like [adui] ‘he hit’ from /aduyi/. The second is (rare) stem-internal sequences in some pronunciations of *nasááye*, *nasááe* ‘God’, *ring'ó'ááni* ‘crested crane’ (sometimes attested as *ring'ó'wááni*) and *ikibi'ráhóóni* or *ikibi'ráóóni* ‘small plate’ – though such sequences may be consistently created from /VyV/ in the perfective stem, for instance *kwaabómóe* ‘we destroyed’ instead of *kwaabómóye* (or *kwaabómóri*). The third source of vowel sequences, which is very com-

mon, arises at the phrasal level with the combination of a word-final vowel before a word-initial vowel. Some speakers (in the available data) simply maintain vowels in hiatus; others (especially EM) seem to at least phonetically systematically merge such vowel sequences into a single syllable, often deleting the first vowel. Tonal evidence supports the conclusion that the two vowels are merged into a single syllable (the phonological tonal facts will be discussed in later chapters). In the transcriptions presented here, a breve mark may be used to notate this property.

[bk]maháraambá ĩsiríini	‘20 wasps’
[rl]rúómö ndara	‘1 room’
[em]mká'rájí öm'táámbr	‘tall judge’
[em]vór-iĩng'inga	‘every moment’
[ml]mboozó'ávó	‘their sister’
[ro]káá'yóóng-önguzúzu	‘young weed’

The breve mark is one way of notating the impression that the two vowel are “run together” into one syllable. The mark might appear on the first vowel or the second, with no implication that there is a systematic difference between first-V reduction and second-V reduction. Similarly, such vowels in sequence might be run together with no space, or there could be a separating space (as observed in the preceding examples); or, there could be a hyphen separating the words (see also *gomée'r-íógónéne* ‘big ship’ with no breve). It was eventually decided that spellings like *gomée'r-íógónéne* should be conventionally used to notate apparent syllabic merger. Location or presence of a breve, or a hyphen, in a vowel cluster simply indicates that the vowels in a sequence phonetically merge into one syllable. The evidence that there is phonological merger pertains to the tone system. As you can see from the variant ways of writing such sequences in the examples above, transcriptions were not systematically and post-hoc regularized to conform to a single convention of spelling. The most-common practice in my data-recording is simply to mark phrasal vowel mergers with a hyphen before the first vowel.

Other variable phonetic details may be retained in the transcriptions. For example there is velarization in some tokens, thus phonemic *mbwá* may be attested as [ɪmbʷuá] ‘dog’, *mwaakweeya* as [mʷaakweeya] ‘2p swept’. Again, the transcriptions have not been filtered to eliminate such variations and low-level detail when they were recorded, on the other hand since they are common, they are not generally recorded in the written versions. The forms [mwaakweeya] and [mʷaakweeya] are phonologically identical; the transcription [mʷaakweeya] indicates that velarization in that token was noticeable, and the transcription [mwaakweeya] either means that there was no noticeable velarization, or that it wasn’t considered important enough at the time to write it (or, it was ignored as well-understood and predictable). Typically, such allophonic details (which are commented on in appropriate parts of this chapter) were not systematically noted in transcriptions, except when they are initially observed, or where they are otherwise “striking” (e.g. uncommon for a particular speaker). Resolving details about these kind of phonetic variants will require a focused and controlled sociophonetic study.

2. Segmental inventory

2.1. Consonants

The clearly-contrastive consonants of Logoori are as follows.¹

	t	[tʃ] ch	k
b	d	[dʒ] j	g
f	s	[ʃ] sh	
v	z		
m	n	[ɲ, ɳ] ɲ	[ŋ] ng'
w	r, l	[j] y	h

The consonant [f] is infrequent, and generally only appears in loan words or before *u*, *w* – however [fwV] is widely replaced with [fV]. Some speakers maintain [fw] in *ífwééza* ‘silver’ vs. [f] in *mféneesi* ‘jackfruit’, but others employ *f* in both words – *ífééza* ‘silver’, *omfenesi*. Both words derive from Swahili *f* (*fedha*, *mfenesi*), and the contrast in Logoori is quite marginal, originating from the fact that *f* originally comes from a voiceless consonant before degree-1 round vocoids, i.e. *u*, *w*. *rífwá* ~ *irífá* ‘thorn’ ← Proto-Bantu **pua*, *rofúro* ‘foam’ ← **pudú*. Transformation of loanword *f* to *fw* as in *ífwééza* is uncommon and probably indicates that the word is an older loan. I have not observed inserted *w* for any speaker with recent words like *ofisá* ‘officer’, never **ofwisá*. Such less-nativized loan words constitute the prima facie case for a contrast between “fw” and “f”. There is generally a degree of uncertainty as to whether a particular instance of *f* has some lip protrusion which might be represented as [fw] or [fʷ], hence a token transcribed as *irífá* probably has a small degree of lip protrusion.

The consonant *b* is somewhat infrequent unless it is preceded by a nasal ([mb] is very common, being the surface result of combining a nasal plus various consonants). Words with bare [b] are often loans.

ebééi	‘price’
ibáága	‘bag’
ibáákuuri	‘bowl’
itáábu	‘trouble’
kebóóko	‘hippo’
kítábu	‘book’
rijííbu	‘answer’

Few words universally have [p]. Candidates would be loanwords with [p] in the source language. A common but not universal strategy for source-language [p] is to voice [p] into [b] when using such a word in Logoori. Some speakers use [p] in these words, some of the time.

¹ The orthographic conventions used here are given in this table, along with the corresponding IPA symbol in square brackets, when they differ.

ibíícha (ipíícha)	‘picture’
ibirabirí (ipiripirí)	‘pepper’
ebóósta (epóósta)	‘post office’
ibúúnda (ipúúnda)	‘donkey’

There are a few words with [p] which have not been observed to undergo voicing with any speaker, so far.²

rí ¹ péera	‘guava’
m ¹ páángo	‘plan’
m ¹ píira	‘ball’

There are also words with [b] which do not appear to vary with [p].³

kíkábo (Sw. <i>kikapu</i>)	‘basket’
msííbi (Sw. <i>mshipi</i>)	‘belt’
abáchi (Luo <i>abaki</i>)	‘abachi tree’ (Uganda Greenheart)
kibáanji	‘water pot’
kiburuburu	‘butterfly’
rí ¹ bwóoni (Luo <i>rabuon?</i>)	‘potato’

There is also variation between [k,g] and [t,d] associated with loan words, such as [ɪgánisa ~ ikánisa] ‘church’ from Swahili *kanisa*, or [ɪdágíga ~ ɪdákíka] ‘minute’ from Swahili *dakika*; [riká¹rádáasi] ‘paper’ from Swahili *karatasi*, [ryáá¹ngázéedi] ‘newspaper’ from *gazeti*. Replacement of voiceless stops with voiced ones is a common nativization strategy, and a major source of inter-speaker variation in pronunciation.

2.1.1. THE PROBLEM OF R/L

The phonetic consonants *l* and *r* are highly variable. Orthographic practice of written Logoori is not uniform, in that both *r* and *l* may be used, with *r* being less frequent. Outside of the speech of FI, I have not encountered evidence for the orthographic rule promulgated in Godia (use <r> after front vowels), but I have frequently encountered both a lateral approximant [l] and a flap that can be written as [r] (more accurately IPA [ɾ]). The distinction is largely rule-governed ([i,y] may condition immediately preceding [l]), but this varies substantially across speakers. No speaker whom I have encountered systematically uses a singleton [l] comparable to Swahili *l* for all instances of the liquid. Even FI, who generally instantiates the singleton liquid as a lateral, pronounces it as [ɾ] (which sounds similar to *r*) and not [l]. Both *r* and *l* are employed in my transcriptions, the choice being based on my judgment of whether the token was more *r*-like or more *l*-like.

² I have not systematically pursued investigation of adoption of Swahili words into Logoori, so these examples arose in the course of ordinary elicitation.

³ On occasion, the word [ɪbárási] ‘horse’ has been gathered in the form [ɪfárási] which is essentially the Swahili word and the immediate source of [ɪbárási], but since this is corrected to [ɪbárási], I conclude that [ɪfárási] is not actually part of Logoori. However, it is plausible that such a replacement has or will take place for some speakers.

Such judgments largely fell out of the transcriptional record over time, since it became clear that more sophisticated analytic methods are necessary to capture the extant range of pronunciations. All instances of <r> could be spelled with <l>, or vice versa, without merging significant general contrasts of the language.

For most speakers, /rVr/ reduces to [ll] (hence *llógoori* ‘Logoori language’ ← /rV-rógoori/, vs. *mrógoori* ‘Logoori person’ ← /mV-rógoori/). The qualitative difference between [ll] and [r] is very significant, and I always write [ll] although “rr” could, by rule, be interpreted as always being pronounced [ll]. The duration of derived [ll] is, at least in certain contexts, typical of that found in various languages which have phonemic geminate [l]⁴ and for all speakers, it is quite evident when ll is word-initial (it can bear contrastive tone: [l¹léési] ‘cloud’). For some speakers, the duration of intervocalic ll is not consistently different from that of singleton l (as observed in their productions of Swahili words). In some instances, such as the verb *kuhólla* ‘to hear’ from /kV-hór-ir-a/, ll tends to sound somewhat long, but sometimes does not appear to be very long.

Especially among younger speakers, /l/ may be emerging as a phoneme which is distinct from, though phonologically related to, /r/. There are some lexical items, all adaptations of Swahili l or English, which for some speakers have non-geminate [l]. For example the adjective *kamili* ‘proper’ always has a short consonant (though it may also be pronounced by some with r), and the word ‘bell’ from Swahili *kengele* may be pronounced *ekéengele*.⁵ The word for ‘Luo (person)’ is always pronounced *mjálwo* ~ *mojálwo*, never *[mojárwo]. The most wide-spread strategy for loanword adaptation and the strategy exemplified in long-established loans is to change Swahili l into r. The innovative strategy exemplified by words like *kamili* is to adopt l as short [l]. As far as can be determined, there is no tendency to adopt foreign l as [ll], that is, delateralization (not gemination) is the only alternative to direct incorporation of [l] from another language.

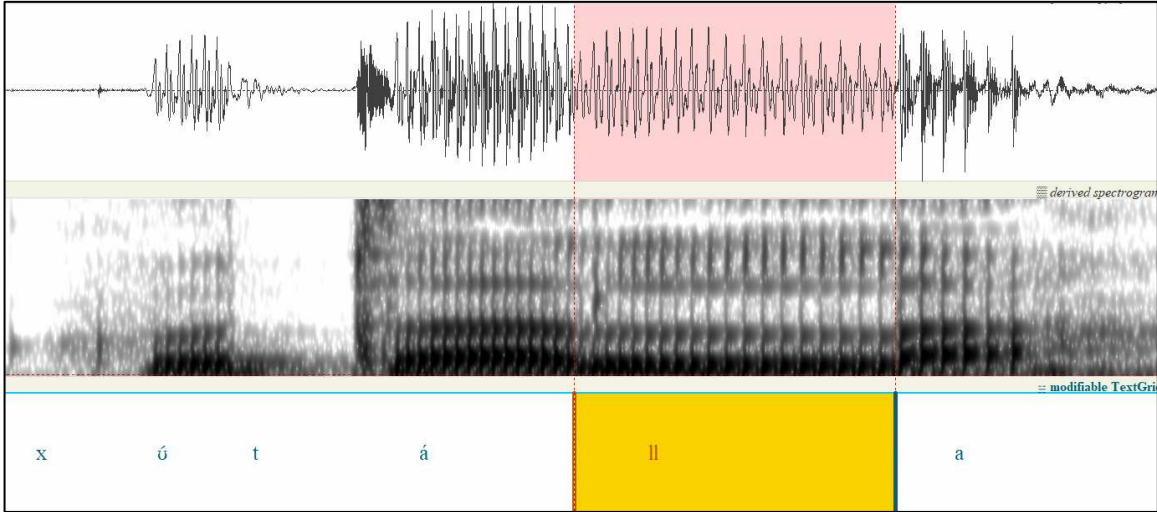
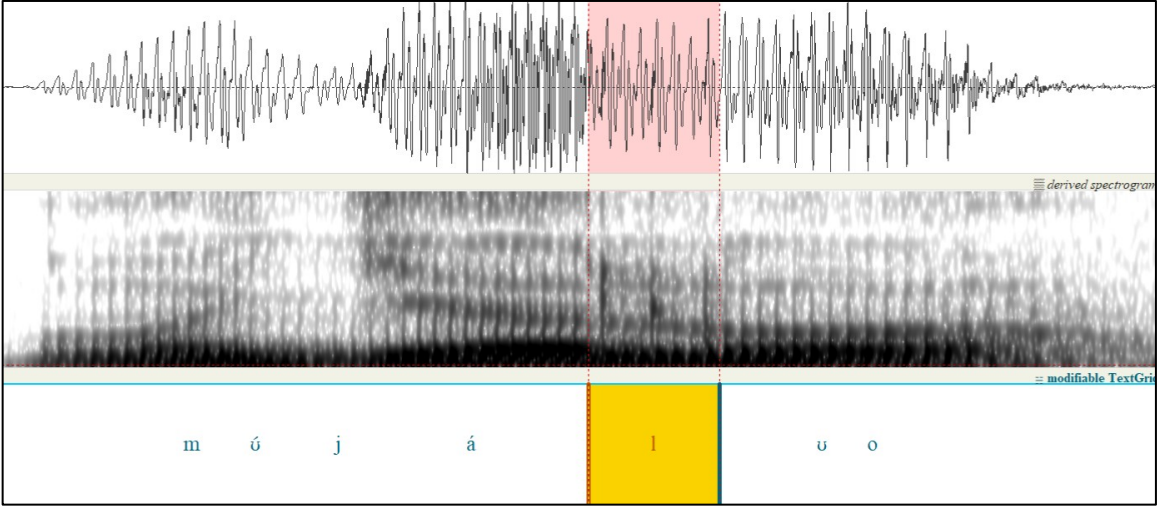
The short version of r is a lingual flap (written here as r) and varies between [r] and a lateral flap [l] or as [l], with both contextual and individual factors being relevant. Instances of [l] in the speech of EM are relatively infrequent compared to BK and RL. Word-initial position and a following i, y are factors that especially favor the lateral flap. Given the wide range of phonetic realizations of the Logoori liquid, only a two-way distinction is made here between <l> and <r> (the former possibly being geminated), reflecting a judgment as being more like canonical [l] versus more like [r] (never trilled [r]).

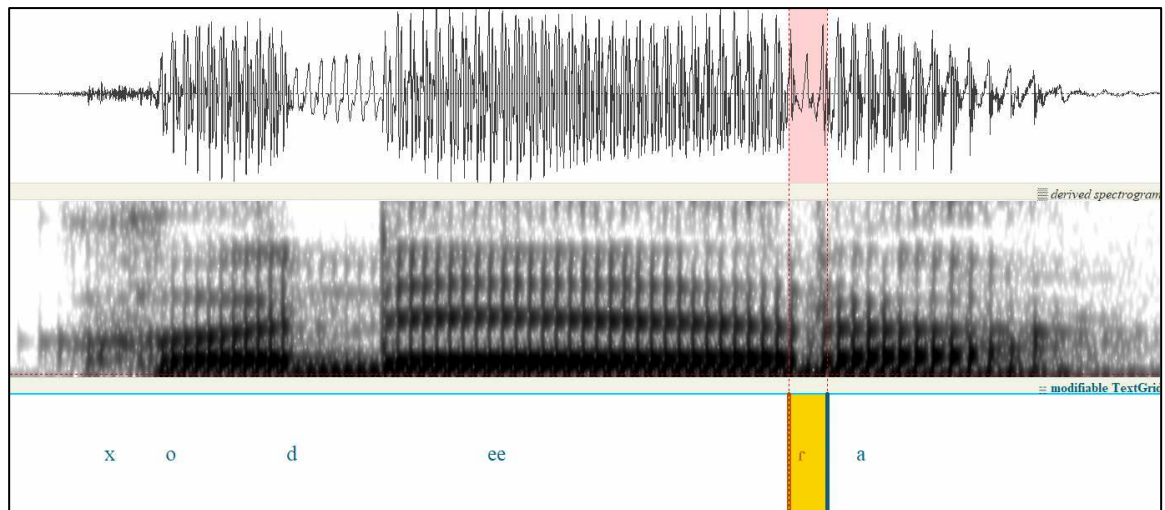
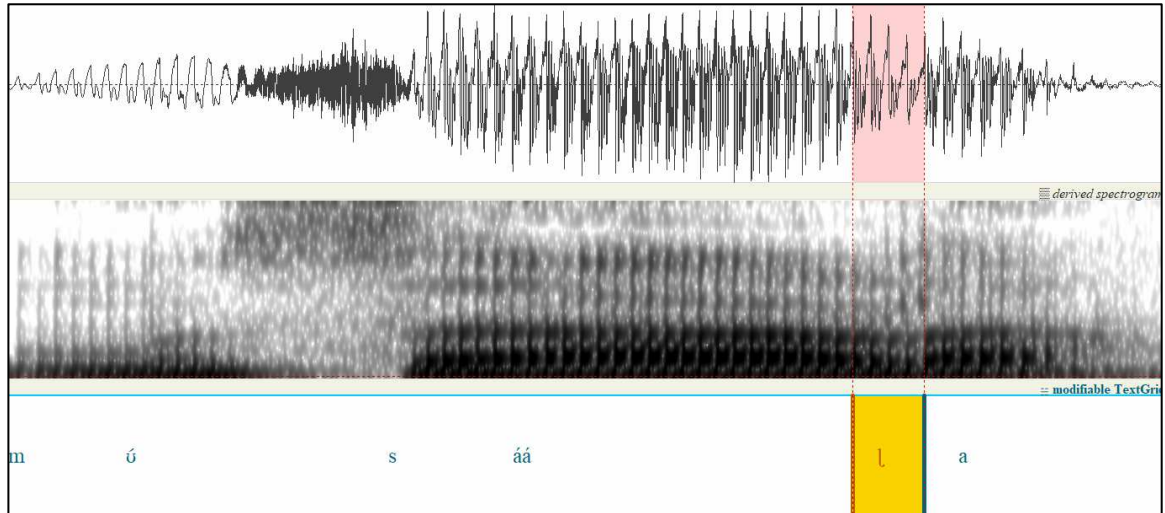
By way of more concrete phonetic detail, below we see four spectrograms of liquids from FI: [mójálwo] ‘Luo’, [xótalla]⁶ ‘to pass rumors for’, [mósáa] ‘tree’ and [xodeera] ‘to treat a person well’. We see a small reduction in amplitude during [l] and [ll], the most obvious difference between the sounds being duration (87 msc versus 210 msc).

⁴ E.g. Saami, Norwegian, Italian, Arabic, Somali.

⁵ The most-nativized form is *egéengere*, which can also be pronounced *ekéengere*. I have so far not encountered **egéengele* with nativization of just the initial consonant.

⁶ FI also predominantly produces velar semi-fricatives – not clear plosives, also not like [x] in German, Russian etc, notates as x in the spectrograms.



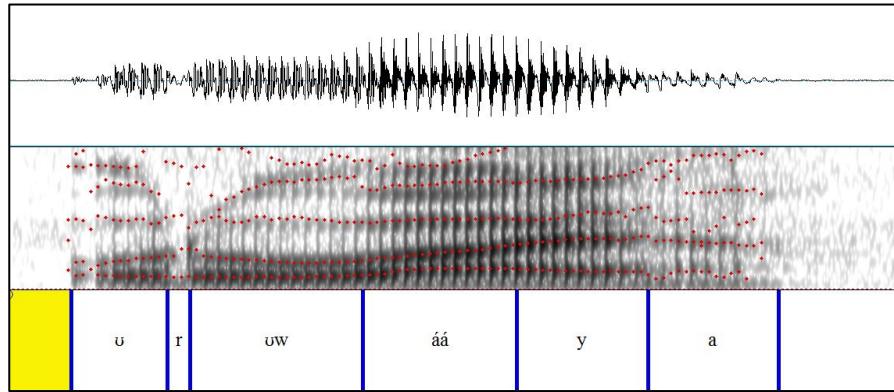


As expected, geminate *ll* is the longest, followed by canonical [l] (in loans). Both [l] and [r] are even shorter – 44 msc vs. 26 msc. Additionally, [r] has a much more noticeable reduction in amplitude, whereas it is very difficult to parse out [l] from the surrounding vowels.

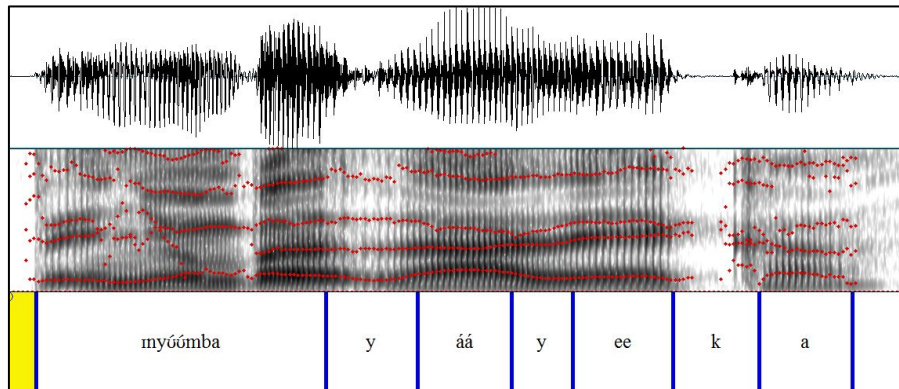
2.1.2. OTHER CONSONANTS

The voiced labial fricative, written as *v* (standard spelling), is somewhere between IPA [β] and [v], except that for FI it is closest to IPA [v] and sometimes [w].

The glide *y* in Logoori, in many idiolects, is phonetically unusual, compared to the segment transcribed [j] in IPA which found in many other languages (such as English, German, Swahili), indeed I know of a parallel only in the Central Kenyan Bantu language Kamba. The front glide need not have a raised tongue position as typically expected from other languages, so it is not always a palatal, it may be a dental approximant. The articulatory essence of this as well as the “dental” *ɲ* pronunciation is a flat tongue configuration, rather than an arched tongue. Leung also notes this feature and transcribes *y* as <ɣ>, referring to it as a ‘dental glide’.⁷ The glide is palatal (which, for clarity, I indicate in this section as [y^j]) when it stands before the tense vowels [i,u], also when preceded by a consonant, and it is realized as a dental glide when syllable-initial before [ɪ e a o ʊ]. For some speakers, the glide is palatal on some (most) tokens of the word *rowááya* ‘wire’ – [rowááy^ja] – a loanword, which allows a comparison of the palatal vs. dental versions of *y* in reasonably comparable contexts. As can be seen in [ʊrowááy^ja], we find the expected shift in formants for [y^j], but no appreciable change in waveform amplitude at that point.



In [inyóómba yááy^jeeká] ‘sagging house’ below, dental *y* has a significant decrease in waveform amplitude during its articulation, reflecting the effect of the narrower approximation.⁸



⁷ Mould 1976 likewise marks *y* with the dental diacritic, but does not comment on that property.

⁸ In this token, the second *y* has some but not as much constriction.

For those speakers with the dental allophone, alternations within a morpheme can be observed depending on phonetic context, for example *kovaaya* ‘to visit’, *avááyí* ‘he visited’. Since this feature is entirely transparent, it is not otherwise noted in transcriptions outside this section. It should also be noted that *y* is often not very distinct before [i]. There are very few contests illustrating [y^hu], but the palatal variant can be seen in [voyúúsi] ‘corn silk’ or [kigá’yú] ‘despised (cl. 7)’.

A related phenomenon is the pronunciation of *n*. Especially among the majority of speakers with the dental glide pronunciation of *y*, the palatal nasal is phonetically dental [ɲ], again with a flat tongue, whereas /n/ is more apical with the tongue tip making contact with the alveolar ridge. There is an alternation for these speakers where /n/ becomes [ny] before *i* and *u*, for example *kokóona* [kokóɲa] ‘to help’, *vaakóonyi* [vaakóoni⁹] ‘they have helped’. The phonological relationship of *ny* and *n* is dealt in more detail in X. Ultimately this contrast derives from an earlier contrast [ñ] versus [ññ] – here using [ñ] to represent the standard IPA palatal nasal rather than the present use of <ɲ> as a variable-nasal phoneme. Earlier singleton [ñ] developed the flat-tongue articulation except before the highest vowels, but geminate [ññ] did not: and then [ññ] was widely degeminated to what is now uniform [ny].

Another source of the surface distinction between [ɲ] and [ny] is the reduction of /n,ɲ/ plus causative /iz/, which creates alternations such as [kohóna] ‘to get well’, [kohónya] ‘to heal (trans.)’, or [kokóona] ‘to held’, [kokóonya] ‘to make help’. The high lax vowels [i, u] constitute a middle ground for lexicalization, since there are tautomorphic sequences [ɲi, ɲu] where one might expect [ɲ], for example *konyira* ‘to stretch’, *myóómba* ‘house’, cf. *koyíimbira* ‘to sing for me’; *nombákáa* ‘I am building’ with the dental realization. The difference is that *ny* in ‘to stretch’ is entirely within the verb root ‘stretch’, and *nyu* in ‘house’ is entirely within the root, at least synchronically, but *n*+{i,u} in *koyíimbira* ‘to sing for me’; *nombákáa* ‘I am building’ comes from the combination of /N/ plus root-initial /y/ (which in these instances is inserted by rule).

The synchronic distribution of [ny] versus [ɲ] can almost be predicted either by identifying bisegmental sources (/kú-hón-ɲ-a/) or else the following vowel (/ɲ/ → [ny] / __i,u), there are unpredictable cases of surface [ny] that have to be lexically recorded – *amá’nyáanga* ‘shakers’, *ovó’nyéégéri* ‘itch’, *irichí’nyéeri* ‘hyrax’, which may be occasionally pronounced with [ɲ]. There are also two problematic verb roots, /nia/ ‘defecate’ and /niaara/ ‘wither, become thin’, where the nasal is followed by a very short high front vocoid that might be mistaken for [y], but this sequence remains phonetically distinct from [ny] as attested elsewhere in the language. These roots will be written here as [nia, niaara]. In short, even though there is a relationship between *ny* and *ɲ* (they are indeed spelled identically in popular writing, as <ny>), the surface distinct sounds will be recorded as pronounced, where <ɲ> is the generally-dental allophone and <ny> is the gen-

⁹ Alternative transcriptions would be [vaakóonji, vaakóopi, vaakóonji, vaakóonji]. What is auditorily distinctive about the “ny” pronunciation is its palatal offglide, and the status of the on-glide from the vowel is not at all distinct,

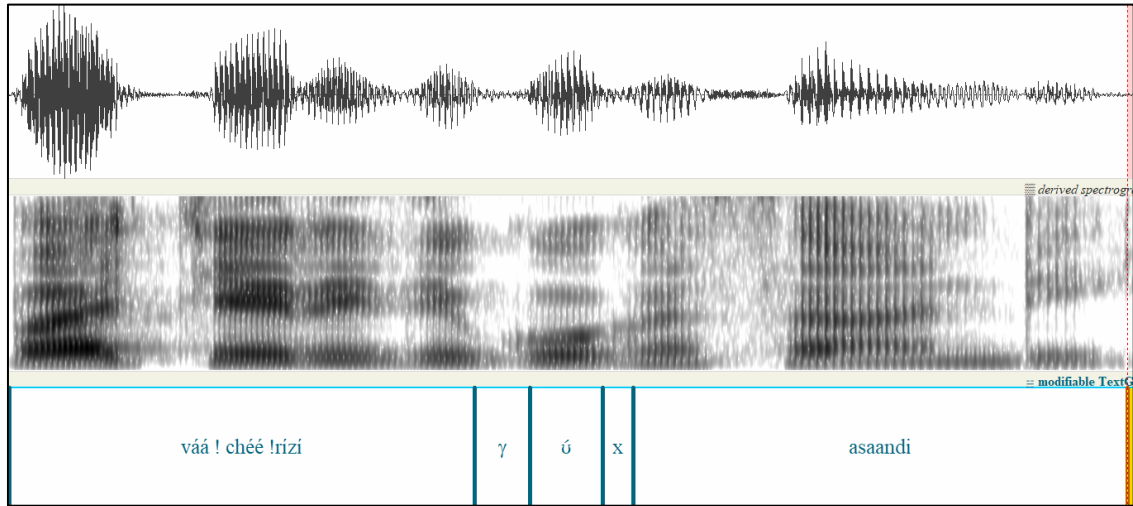
erally-palatal allophone – and where some speakers simply pronounce <ɲ> the same as <ny>.

Data from FI points to a better understanding of the historical antecedent of the variants of “ny”. He does not evince the dentalization phenomenon discussed above, and his pronunciation of “meat” is (IPA) [ɪɲama] not [ɪɲama]. Cases of surface [ɲj] from other speakers, for example [kɔgavɔrannya] ‘to divide up’, have a geminate *n* followed by [y] (alternatively, a geminate palatal nasal), where most other speakers have singleton [ny] for [nny]. A systematic exception is [nnya] ‘mother’ which is apparently not pronounced as *[ɲya] by anyone. Some other speakers occasionally produce geminate [nny] in a stem instead of [ny], for example FA *kɔganagannya* ‘to think’, SY *kɔdɪnnya* ‘to be hard’, BA *kɔbiɪnnya* ‘to press’. We may then presume that original *nny* was shortened for some speakers – the majority of those in my data, but the distribution on the general Logoori-speaking population has to be studied independently.

The fricative *sh* ([ʃ]) is highly variable. The original situation is that there was no [ʃ], and there was a contrast between [hy] and [sy]. Palatalization of *hy* to *sh* is quite widespread, compared to palatalization of *sy* to *sh*. A few speakers have been identified who synchronically maintain a three-way distinction between *sh*, *hy* and *sy*, though for those speakers, robust [ʃ] is only found in loan words such as *ɪmɪʃʃeni* ‘mission’, *riʃʃáhídi* ‘witness’ (from Swahili), and *hy* seems to vary freely with *sh*. Most speakers uniformly neutralize *sh* and *hy* to *sh*, thus *kɔhyóósha* ~ *kɔshóósha* ‘to be warm’. Speaker LI tends to preserve a distinction between original *sy* and *hy*, the latter generally being pronounced as [ç] e.g. *kóʃá* ‘to be warm’. Finally, some speakers systematically have *sy* where other speakers use *sh*, e.g. *kɔsyeeena* ~ *kɔsheena* ‘to step’. EM, for example, uses *sh* in these words, whereas BK uses *sh* in the former example and *sy* in the latter, and PM uses *hy* and *sy* as well as *sh* (at least potentially). An additional context for deriving *sh* is that /h/ becomes [ʃ] optionally or obligatorily (depending on speaker) before suffixal *i*, for example *kwááha* ‘to pluck’, *nzahi* ~ *nzashi* (EM) ‘I plucked’. Paradigmatic fronting of *h* before *i* is much more variable than tautomorphemic pronunciation of *hy*.

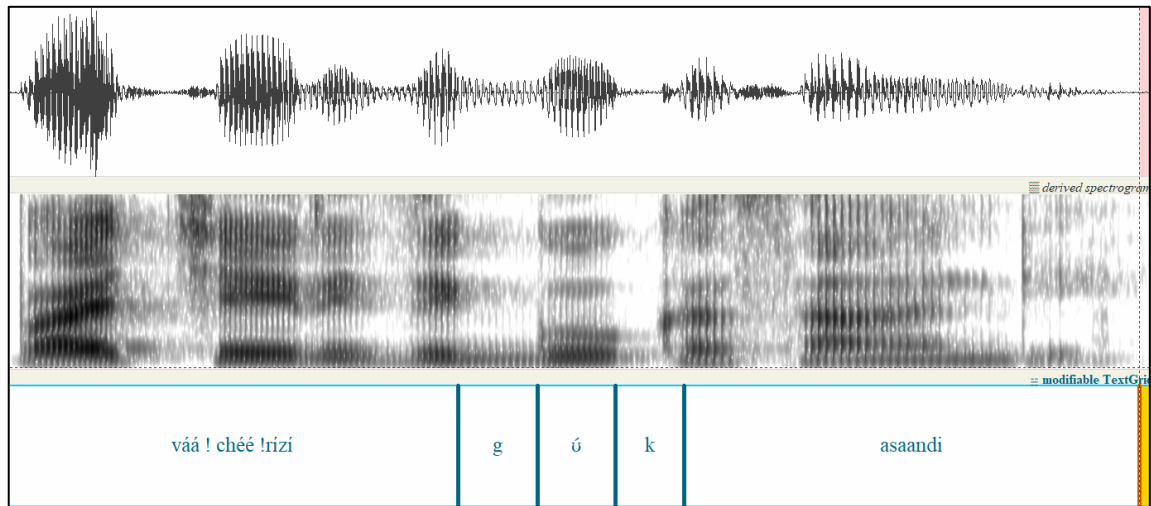
The fricative *z* is widely pronounced as [z], but some speakers (esp. BK, PM and FI, the latter being more systematic in using that pronunciation) optionally pronounce this as an affricate [dz], with a relatively short closure period.

The two velar phonemes /k, g/ are treated here as stops however they are sometimes or even often realized as (quasi)-fricatives. For the most part, /k/ is phonetically [k], but a notable exception is that FI produces a significantly lenited version of that sound, moreover, other speakers may attest such options under certain conditions. The voiced obstruent /g/ is lenited to some extent by many, perhaps most, speakers. We consider below two consecutive utterances of *vááʰchééʰrízi gó kasaandi* ‘they certainly did greet Kasaandi’ produced by EM. It is apparent that after the first utterance he deemed his pronunciation to be sub-optimal (speakers *can* actually monitor their phonetic outputs, they are not blind to phonetic variation as is sometimes supposed). The crucial difference between the utterances is that in the first, faster performance, /k/ and /g/ are lenited to fricatives.



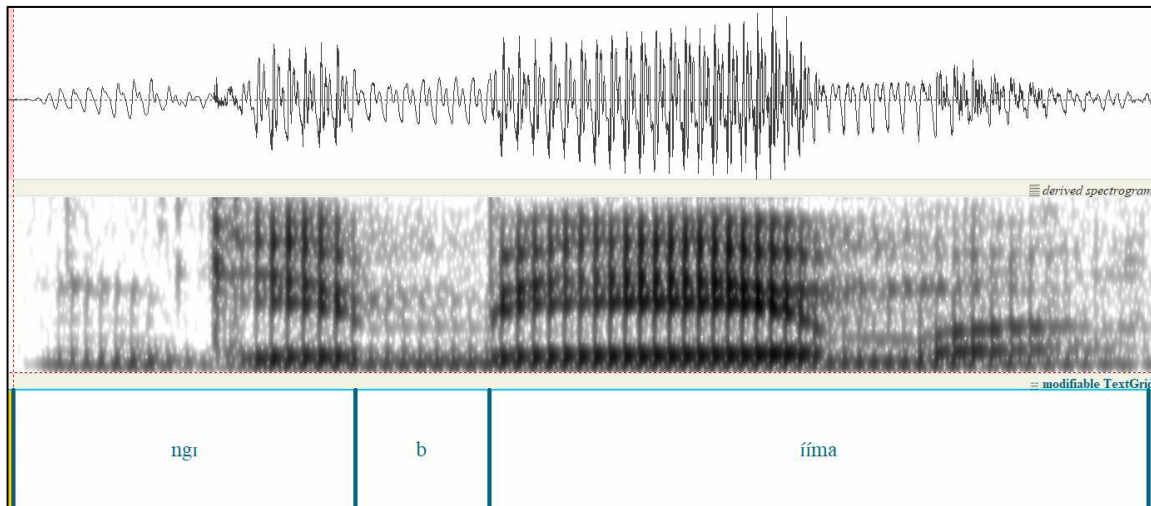
We observe that the margins of *k*, *g* here are not very sharp and there is nothing resembling a release burst.

In the immediately-following performance, the margins of /*k*, *g*/ are much sharper and there is a clear release burst in both cases.

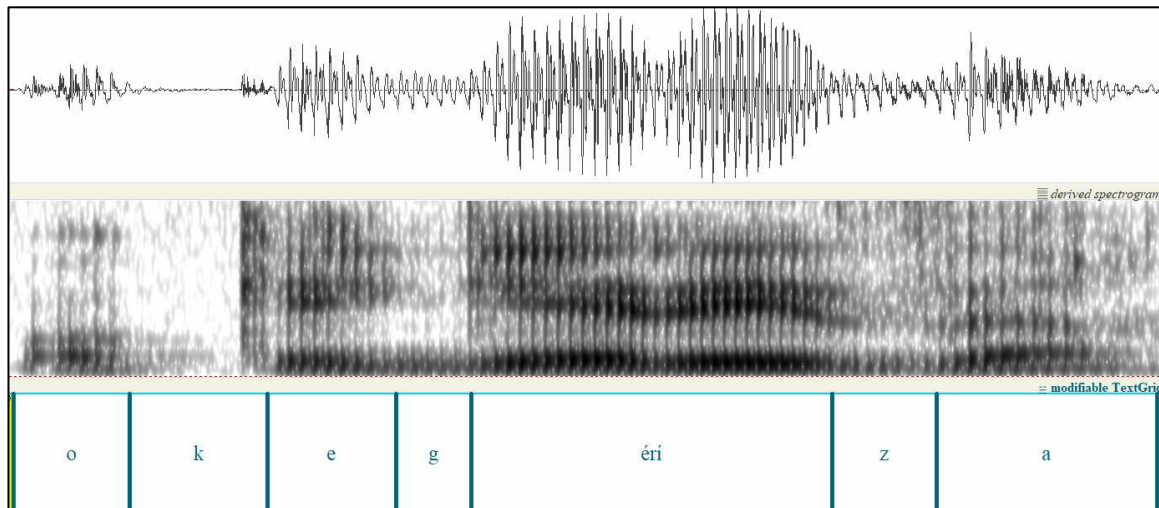


There is a difference in duration of these consonants in the two utterances – $\gamma = 68$ msc, $x = 37$ msc and utterance duration = 1353 msc in the first token, but $g = 107$ msc and $k = 92$ msc with utterance duration = 1474 msc in the second utterance. Almost all of the difference in duration between the tokens is attributable to the faster (lenited) vs. slower (fortis) productions of the velars.

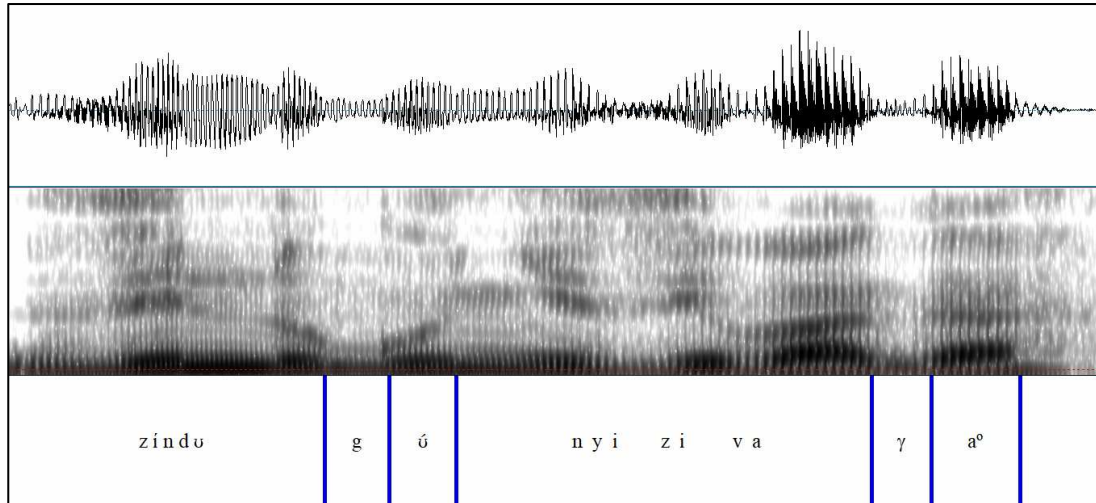
The spectrograms of *ngibiíma* ‘I am still measuring’ and *okegériza* ‘you are still trying’ provide a baseline for distinguishing stops and fricatives. In *ngibiíma*, we see a reasonably clear and abrupt distinction between the stop *b* and the surrounding vowel.



In *okegériza* we see a similar distinction between the surrounding vowels and the consonant *g*, with the onset of *g* being more-blended with the preceding vowel. By contrast, it is hard to draw a clean line separating *z* from the surrounding vowels, and amplitude during the stops *g*, *b* is much lower than during *z* (because the stops have greater closure).

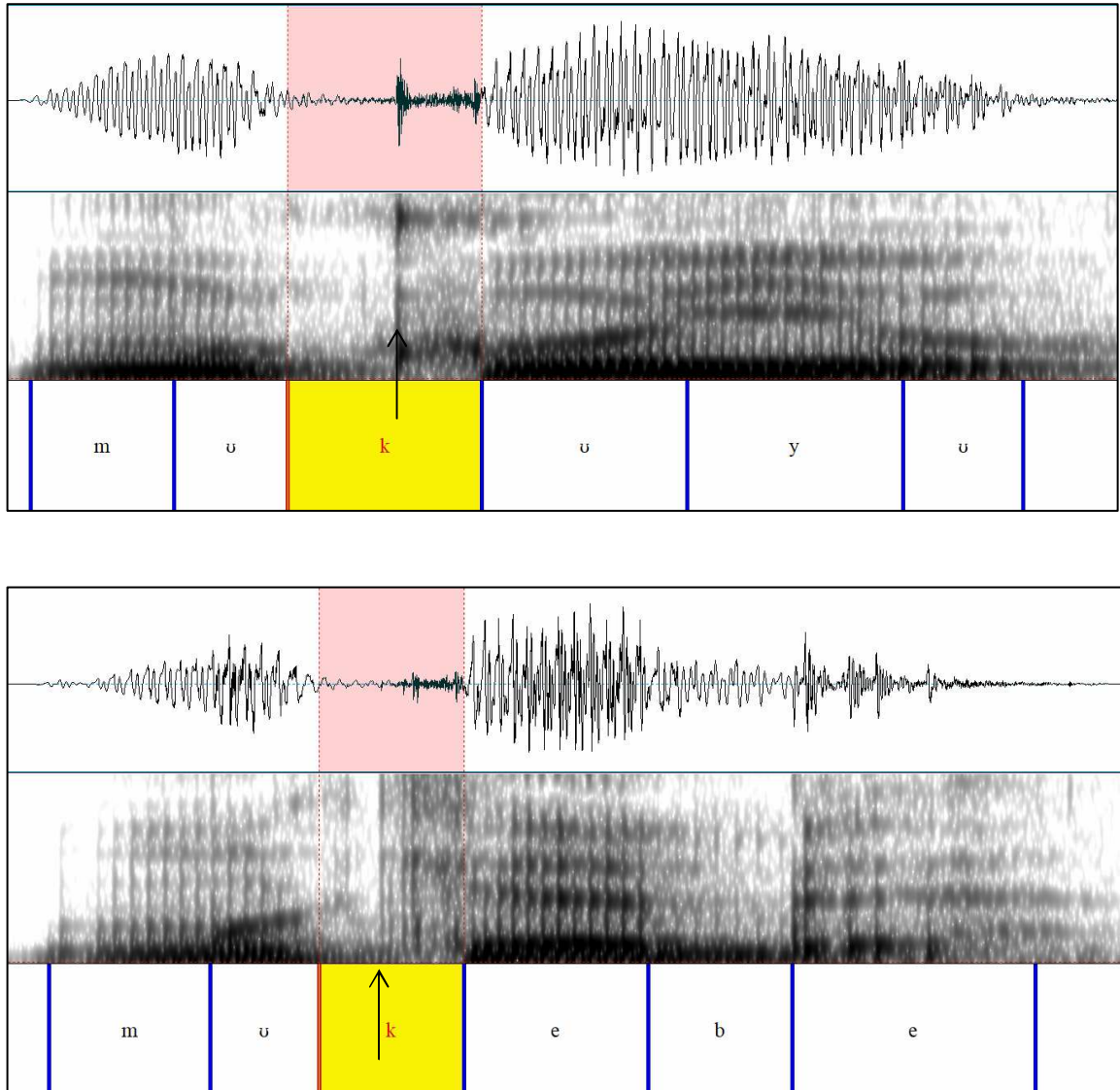


All speakers seem to weaken /*g*/ to some extent. Another example from LI is [zí'ndógúnyi zivaya°] '3 ants' where /*g*/ is realized as a stop in the noun [zí'ndógúnyi] but as a fricative in the numeral [zivaya]. The segmental margins of /*g*/ are clearer in the first instance and there is a visible release burst, but in the second instance voicing during the consonant is attenuated, and the waveform is more irregular in shape.



FI produces a lenited version of /g/ most of the time, even after a nasal.

It was noted previously that [x] is said (per the AIM grammar) to exist in some dialects, which is the case for speaker FI although it is not a full fricative as in Tiriki, it is a phonetically lenited stop. The percept [x] is most prominent utterance-initially, to the point that all instances of /k/ in that position from other speakers sound like [x]. This is because utterance-initially, there is no acoustic consequence of the stop closure, all that one hears is the fricative-like release. Intervocally, there is a degree of variation, so that (speaking in approximate terms) about half of the time the voiceless velar sounds like [x], otherwise [k] or [kx] – aided by the fact that the stop closure can be detected from its difference from the preceding vowel. Consider the following two tokens from FI, [mukuyo] ‘fig’ and [mokebe] ‘tin’. In the former, there is a clear release burst within *k*, which is not present in the later token. The transition from closure to fricative opening is indicated by an arrow in the transcription: it should be noticed that the presumed closure is not acoustically silent (we also see from the margins that this is not environmental noise). It is conjectured that the closure is brief and often partial, so that there is some acoustic ‘leakage’ from the preceding vowel into the closure of /k/. In other words, /k/ lenites to a “weakened k”, not being a full fricative nor a full stop.



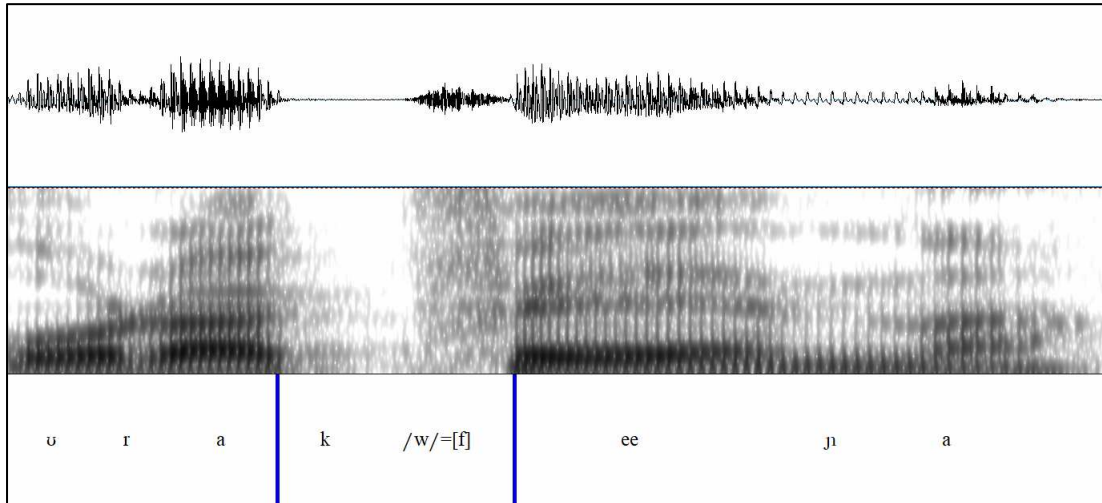
Another variable phonetic detail is that the clusters *mw*, *bw*, *vw* may be pronounced with fricativization and unrounding of the glide, for example $_{[bk]}mbwá$ $[mb^v\underset{H}{u}á]$ ‘dog’ $_{[fa]}mwaakweeya$ $[m^v\underset{H}{a}akweeya]$ ‘2p have swept’, $_{[rl]}mwaaráa$ $[m^v\underset{H}{a}aráa]$ ‘2p are spreading’, $_{[em]}v\underset{H}{v}wóóngo$ $[v\underset{H}{v}wóóngo]$ ‘brain’.¹⁰ This phonetic process is most noticeable in the case of */mw/*, where */w/* may delete without a trace in some tokens, e.g. $_{[rl]}[umíívi\underset{H}{u}mdáámaanó]$ ‘bad thief’ alongside $_{[rl]}[umwíívi\underset{H}{u}mdáámaanó]$; $_{[em]}[yaarí\underset{H}{u}mígizi]$ ‘he was a teacher’.¹¹

A similar process affects */kw, gw/* which are realized as $[kp, kpf, gb, gbv, gv]$, for instance $_{[rl]}[umsáara\underset{H}{u}gbaakpáádika]$ ‘a split tree’, $_{[em]}[umpííra\underset{H}{u}gbaakpáádika]$ ‘a burst ball’

¹⁰ This feature is also noted in Mould’s transcriptions $[m\underset{H}{w}]$ for *mw*.

¹¹ There is no practical way to determine if *w* is actually deleted as opposed to being rendered most difficult to detect: by comparison, some evident correspondent of *w* is audible in all tokens of *vw*, *bw*, except in the case of deletion of *w* before *u*, which is an optional rule.

(*gwáákwaádika*), [kpʰáámkoba] ‘we beat him’ (*kwáámkoba*) [yáá¹ngvíira] ‘he fell on me’ (*yaangwíira*); [fá]moryáángo gbííguchi ‘open door’ (*gwííguchi*), [aʌgbíírɪ] ‘he fell on me’ (*aangwíírɪ*); [pm][kpóómbaka] ‘to build’ (*kwóómbaka*), [kugba] ‘to fall’ (*kugwa*). An example of [úrákfēɛpa] = /ʊrakóɛpa/ ‘you will want us’ from LI is seen below, where a clear fricative release is seen after the stop [k].

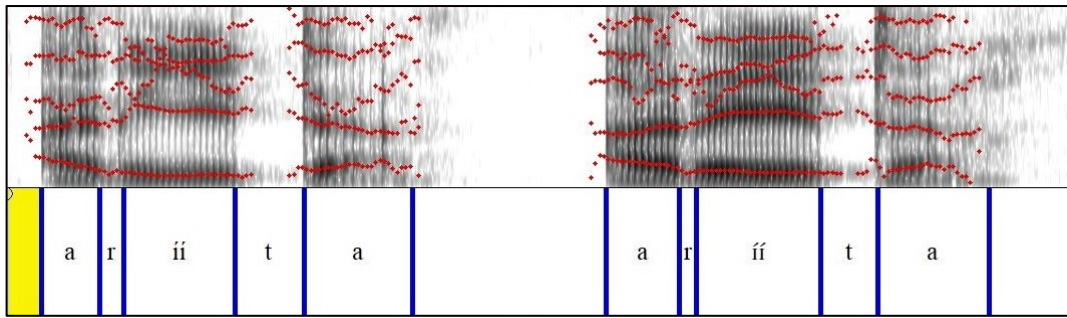


One final pair of consonants should be mentioned, namely [θ, ð] which would conventionally be spelled as <th, dh>, and will be so spelled here on the rare case that they arise. These fricatives arise for some speakers in words taken from Swahili, for example *thamaniini* ‘80’, also pronounced *dhamaniini*, or *féédha* ‘money, silver’, *idháhaab* ‘gold’. These words are usually nativized so that θ→s and ð→z (*dhamaniini* is also recurring in the data for FA). I have not pursued the question of how English [θ, ð] are adapted. It is important to note that these words are still subject to rules of Logoóri phonology such as the regular tone rule discussed in Q generating the contrast between *zisózi dhamaniini* ‘80 fish’ and *zindóóngóózi dhámaniini* ‘80 peaks’, therefore this is not code switching into Swahili. Nevertheless, this is not a general sound of Logoóri, it is analogous to the use of [ø] and [x] in English in “milieu, Bach”, by speakers who know French or German.

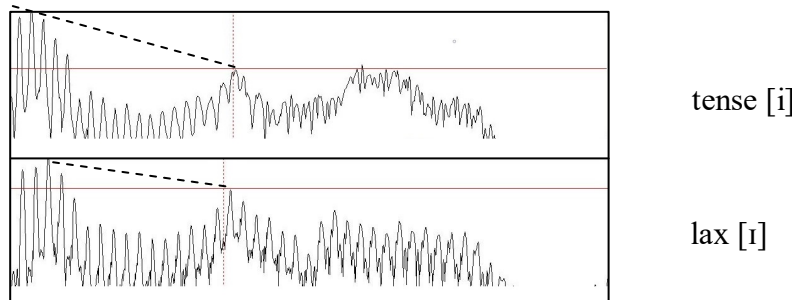
2.2. Vowels

There are 7 lexically underlying vowel qualities in Logoóri: [a e o ɪ ʊ i u]. All vowels may be long or short, long vowels being indicated by doubling the vowel letter, for example *kohana* ‘to close’, *koháana* ‘to give’; *kohéra* ‘to come to an end’, *koheera* ‘to breathe’; *kohólla* ‘to hear’, *kohóulla* ‘to thresh for’.

The vowels [e, o] are comparable to Swahili *e, o* and IPA [ɛ, ɔ]. The distinction between [i, ɪ] and [u, ʊ] is quite subtle. There are very few minimal pairs in word-medial position, e.g. [aríita] ‘he will kill’ and [aríita] ‘he will bury himself’.

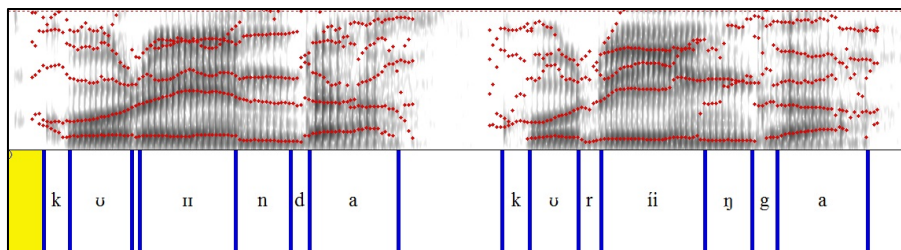


The following spectral slice in the range 0-5500 Hz compares the energy distribution of [ɪ] and [i], where we can see that the tense vowel on the top display has a more pronounced downward spectral tilt compared to the lax vowel on the bottom. Compare the greater decrease in amplitude between F1 and F2 for [i], and the lesser decrease in amplitude between these formants for [ɪ]. This is consistent with the impression that the tense high vowels are somewhat breathy-voiced – the spectrum of breathy vowels has a larger negative slope compared to modal-voiced vowels.

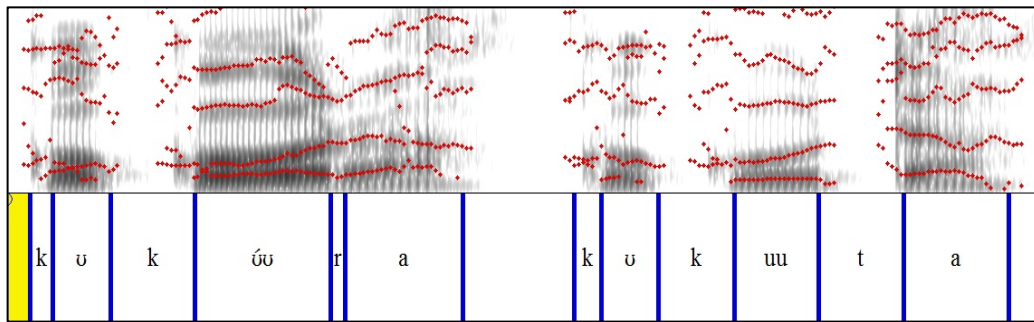


The spectra and spectrograms indicate that the first two formants of [ɪ] are further apart than those of [i], by about 200 Hz.

A general grammatical source of minimal pairs between [ɪ] and [i] is the difference between singular imperative with object prefix (*korɪndi* ‘guard us!’, *gigorɪ* ‘buy it!’) and plural imperative with object prefix (*korɪndi* ‘guard-pl. us!’, *gigorɪ* ‘buy-pl. it!’), with high root vowels. An example of a near-minimal pair *korɪnda* ‘to guard’ vs. *korɪnga* ‘to fold’ is seen below.



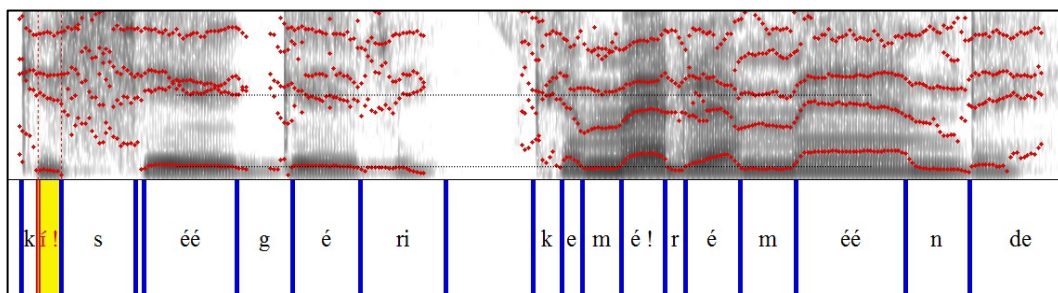
The near-minimal pair *kokóura* ‘to extract’ vs. *kokuuta* ‘to scrape’ are seen here.



The vowels [i,u] tend to partially devoice in favorable environments, which can aid in identifying the vowel distinction.

Although the distinction between [i u] and [ɪ ʊ] is linguistically significant, it is also ‘tenuous’ in that expected [i,u] in a root may be pronounced with [ɪ,ʊ]. This plays a role in lexical variation, where ‘leaf trash’ may be pronounced [amavururi] or [amavorori], and ‘mole’ may be [iʼmbókú] or [ɪʼmbúkú] (even from the same speaker). Certain common roots are consistent, for instance the root *-ndu* ‘person’ in *omvondo* is (probably) never pronounced **-ndu*; the root (orthographic) *-itu* ‘market’ is always pronounced with [ɪ], not [ɪ] as in *ichiitu*. Root-variation between [i,u] and [ɪ,ʊ] is especially common when the following vowel is [i,u], suggesting an earlier or emerging rule of regressive tenseness harmony. As a general rule, [i,u] are much less common compared to [ɪ,ʊ], and a speaker is most likely to produce a borrowed or unfamiliar word having a high vowel with [ɪ, ʊ] rather than [i, u]. The high degree of variation over [i, ɪ] and [u, ʊ] suggests that the distinction may be lost in a few generations.

There is also phonetic raising of the mid vowels *e, o* ([ɛ, ɔ]) which become tense [ɛ̃, ɔ̃] before *i, u*. For many speakers, this process applies iteratively through a string of mid vowels. In the examples below from SY *kíʼsyéégéri* ‘sty’ versus *keméʼréméende* ‘candy’, the lowest two formants are much closer together in the latter (more closely resemble *a*) compared to the former (more closely resembling *i*). Formant comparison lines are aligned to F1 and F2 of *ee* in *kíʼsyéégéri*.



Older speakers tend to have a more “phonologized” version of mid-vowel raising and younger speakers seem to optionally raise vowels only when the mid vowel is right before the high trigger. Some speakers occasionally devoice prepausal *i, u* but not other vowels. Nevertheless, data from speaker EM exemplifying /e,o/ before [a,e] versus [ɪ]

Figure 1 is a vowel plot showing the relationship between F1 and F2-F1 for Tense and Lax variants of mid vowels. The plot includes data points for various vowels (e, ε, ɐ, ɪ, ʊ, ɔ, ɒ) and their tense/lax variants, with horizontal bars indicating the range of F2-F1 values for each vowel class. The x-axis represents F2-F1 (Hz) from 1850 to 250, and the y-axis represents F1 (Hz) from 300 to 750.

Over a corpus of 107 vowels, there was no significant difference in F₂ value as associated with the following vowel context (i.e. /e/ before [i] versus /e/ before [a]), but there was a significant difference in F₁ value. Thus it is credible to claim that there is a raising harmony process in the language affecting mid vowels. Moreover, there was no significant difference in F₁ depending on whether the mid vowel is immediately before [i] versus being separated from [i] by one or more syllables with /e,o/ (which are assumed to harmonize and thus to be the immediate cause of vowel raising in e.g. [kɔ̞vɛ̞zɛ̞gɛ̞rɪ]) – but /e,o/ followed by [aC₀i] is different from [eC₀i]. This latter fact suggests an iterative categorial rule. It is tentatively concluded that there is a late phonological rule tensing /e,o/ to [ɛ ɔ̞] before [i] (and presumably [u], but constructing contexts is difficult) and derived [ɛ ɔ̞]. There are a few cases where [ɛ] appears apart from following [ɔ̞ ɛ i u], the most striking being [ribɛ̞ba] (LI), [iribɛ̞ba] (BA) ‘small tank’ and [esɛ̞twe] ‘mouse bird’ (LI, BA), also [kokɛ̞ryɲa] ‘to gorge’ (BA) with iterative raising, presumable triggered by /ny/. The status of completely unexpected tensing in *iribɛ̞ba*, presumably derived from the Swahili verb *-beba* ‘carry’ is a mystery – such tensing is not a characteris-

tic of Swahili loan words (*ebéde* ‘ring’, *ebéénzeni* ‘wash basin’, *omochéére* ‘rice’). This is also not a wide-spread exceptional tendency. Surface-unpredictable [ɛ ɔ] (not before /i u/) are extremely rare in the data, and no attempt is made to keep track of all tokens which manifest such e-raising.

An exception is that mid vowel raising is robust and plainly phonological in the speech of FI.

kodeng'a	to raise for easy kicking
kodɛnnya	to make raise for easy kicking
kohóna	to get better
kɔhɔnnya	to heal
kóménya	to live
kɔmɛnnya	to make live
kusyeena	to step
kusyeɛnnya	to make step

The trigger for raising here is the causative suffix /iz/, which both triggers mid vowel raising as well as the ng' → ny change, then iz → iny and n(y)iny → nny. Comparable examples from other speakers, such as EM, only show [kohóna, kohónya] and [kusheena, kusheenya].

There is also a tendency for the lax high vowels /ɪ ʊ/ to become tense before [i, u] and [u]. For example, the cl. 7 prefix /kɪ/ may be pronounced as [ki] in [kivúnɪ] ‘reason’. It is not clear whether this tensing results in a categorial neutralization of /i/ and /ɪ/. It is possible that the resulting vowel is just “somewhat tenser”. Such tensing is at any rate optional. There is at least one lexicalized example of clear high vowel tensing, that the root /-ig-/ ‘learn’ has a tense vowel in /-igiz-/ ‘teach’, but not all speakers produce [kwíígiza] rather than [kwíígiza].

3. Prosody: Vowel Length, Tone, Syllables

3.1. Vowel length

Vowel length is significant, and is indicated in some traditional orthographic practices. Nevertheless, perfectly-controlled lexical minimal pairs can only be provided in restricted grammatical contexts which also control tone differences. In a verb, the root would either have to be lexically L toned or have at least two syllables following the long vowel (otherwise the long vowels would differ in tone as in [kodónyé] ‘let us make s.t. into small pieces’, [kodóónyé] ‘let us talk aimlessly’. Near-minimal pairs are seen below.

kuguumara	‘to grow large’	kugumira	‘to catch’
koheera	‘to breathe’	kohéra	‘to come to an end’
kohúulla	‘to beat for’	kohólla	‘to hear’
kukuuva	‘to flatter’	kukóva	‘to be partially cooked’
kopoora	‘to get’	kopóra	‘to strip leaves’
kosáámora	‘to slap’	kosamora	‘to go to work’
kosona	‘to point at’	kosooma	‘to read’

kutóma	‘to send’	kotuuma	‘to jump’
imbíizi	‘warthog’	imbítí	‘hyena’
imbóra	‘rain’	imbóúza	‘strong wind’
mféneesi	‘jackfruit’	mféréji	‘water tap’
mukáádo	‘avocado’	kíkábo	‘basket’
mukóro	‘initiate’	rikóóro	‘pigeon’
eméeri	‘ship’	emére	‘mashed cooked bananas’

Surface distinctions derived by the application of general or construction-specific rules also leads to an opposition between long and short vowels.

vaafúúti	‘they erased me’	vafúúti	‘they erased’
yaakóúsijikiza	‘he has annoyed me’	yaakusijikiza	‘he has annoyed’
aríita	‘he will kill’	arítá	‘he may bury’
varáata	‘they will do surgery’	varata	‘they will bury’;
kodeechi	‘we cooked’	koodéechi	‘we have cooked’

Vowel length is not lexically contrastive before NC clusters, but unlike the situation in some Bantu languages, both long and short vowels are surface-possible before NC.

nzámhááyaa	‘I am swinging’
ɲimbáa	‘I am singing’
ɲómbákáa	‘I am building’
ɲééngaa	‘I am brewing’
ɲííngúkaa	‘I am melting’
ɲóómbóoraa	‘I am over-pouring’
ímbwá	‘dog’
zíímbwá	‘dogs’
ɲji	‘fly’
ziinji	‘flies’

Phonological alternations in vowel length are complex, as discussed in later chapters.

One phonetic tendency is worth noting for posterity. In connected speech, especially on longer utterances where a long vowel appears further to the left, the duration of a long vowel may be reduced compared to what one encounters in the third- or second-from-last syllables. Even in relatively short words, one finds durational variation leading to transcriptional differences like *kwayéyá* ‘if we sweep’, *kwaáyéná* ‘if we want’; *yáyiga* ‘he has learned’, *kwááyiga* ‘we have learned’. When attention is paid to pronunciation, we will find *kwaáyéyá* and *yááyiga*, and short prefix syllable pronunciations are relatively infrequent. In other Bantu languages, this is the historical precursor to loss of vowel length distinctions before the antepenultimate syllables, as exists in Chimiini, Matumbi and Mwera, inter alia. The phonetic seeds of that change are manifested in current Logoori, and may be phonologized in a few generations.

3.2. Tone

Logoori is a tone language, and distinguishes two surface tones, H and L (the former marked with acute accent). Lexical strict minimal pairs are rare.

kwíizoriza	‘to remember’	kwíízoriza	‘to fill’
koviimba	‘to roof’	koviimba	‘to swell’

However, tonal minimal pairs reflecting grammatical differences are frequent in the language.

váámíga	‘they strangled’	váámiga	‘they strangled me’
yáákagura	‘he has bought’	yaakagórá	‘he bought’
umúúndú yááváriza	‘the person counted’		
umúúndu yááváriza	‘the person who counted’		

3.2.1. SYLLABLE DYNAMICS OF TONE

Viewing tone from the syllabic perspective, there are three surface distinctions, High (H), Low (L) and Fall (F) – additionally, there is a general pitch-register lowering process of downstep (notated with ¹) which occurs between H tones. A final surface tone marker is ° which indicates “non-falling”, a rule-governed surface property of some pre-pausal syllables, for instance [na kodeeke°] ‘we will cook’ from /na kodeeké/ – an alternative transcription would be [nā kōdēēkē] with mid tones on all vowels.

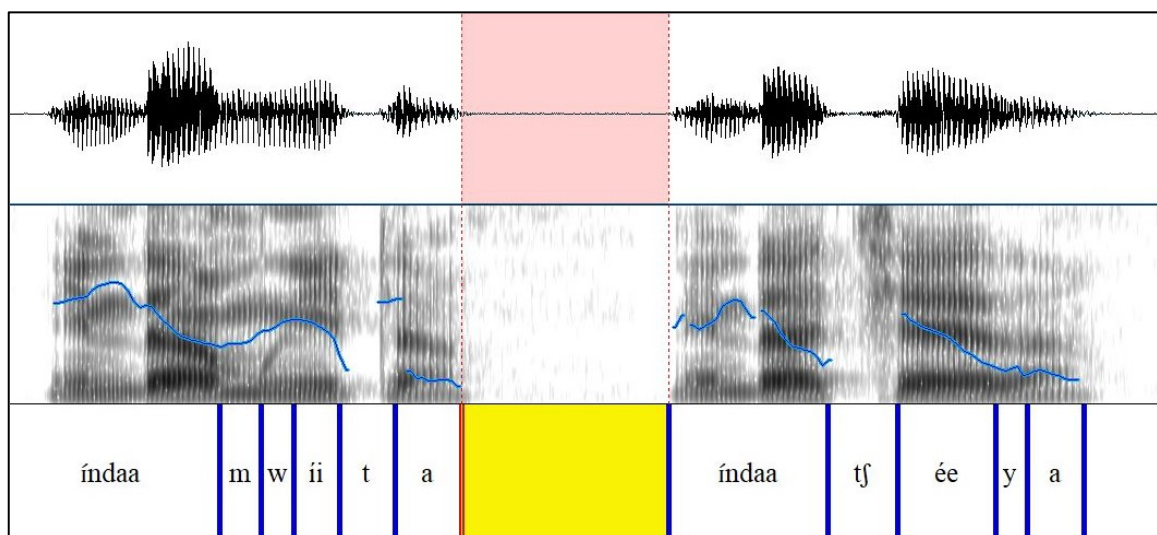
Falling tone only occurs on a long syllable, and is notated as H (accent) on the first vowel, e.g. *kodeéka* ‘to cook’. F is highly restricted. There are grammatical and lexical contrasts between F and H in the penultimate syllable, for example:

ijóombi	‘salt’
éng’éende	‘jigger’
ikíráato	‘shoe’
omosáaza	‘husband’
kodeéka	‘to cook’
ni vakaráange	‘they will fry’
yáakadéeka	‘he has just cooked’
iráánji	‘color’
orobááng’a	‘panga’
ekékóómbe	‘cup’
ipááña	‘tomato’
akedééchi	‘he cooked it’
yáadééka	‘he cooked (remote)’
akeróonda	‘he is dreaming’

It should be noted that penult F is rare in CVVCV nouns, compared to level H. In verbs, the contrast is governed by grammatical tone pattern. There are two types of fall, one being H-to-L fall, the other being H-to-¹H fall, seen in [egó¹ófyá] ‘hat’. This (infrequent) H-to-H fall generally derives from CVVCV̌, and alternates with CVV¹CV̌.

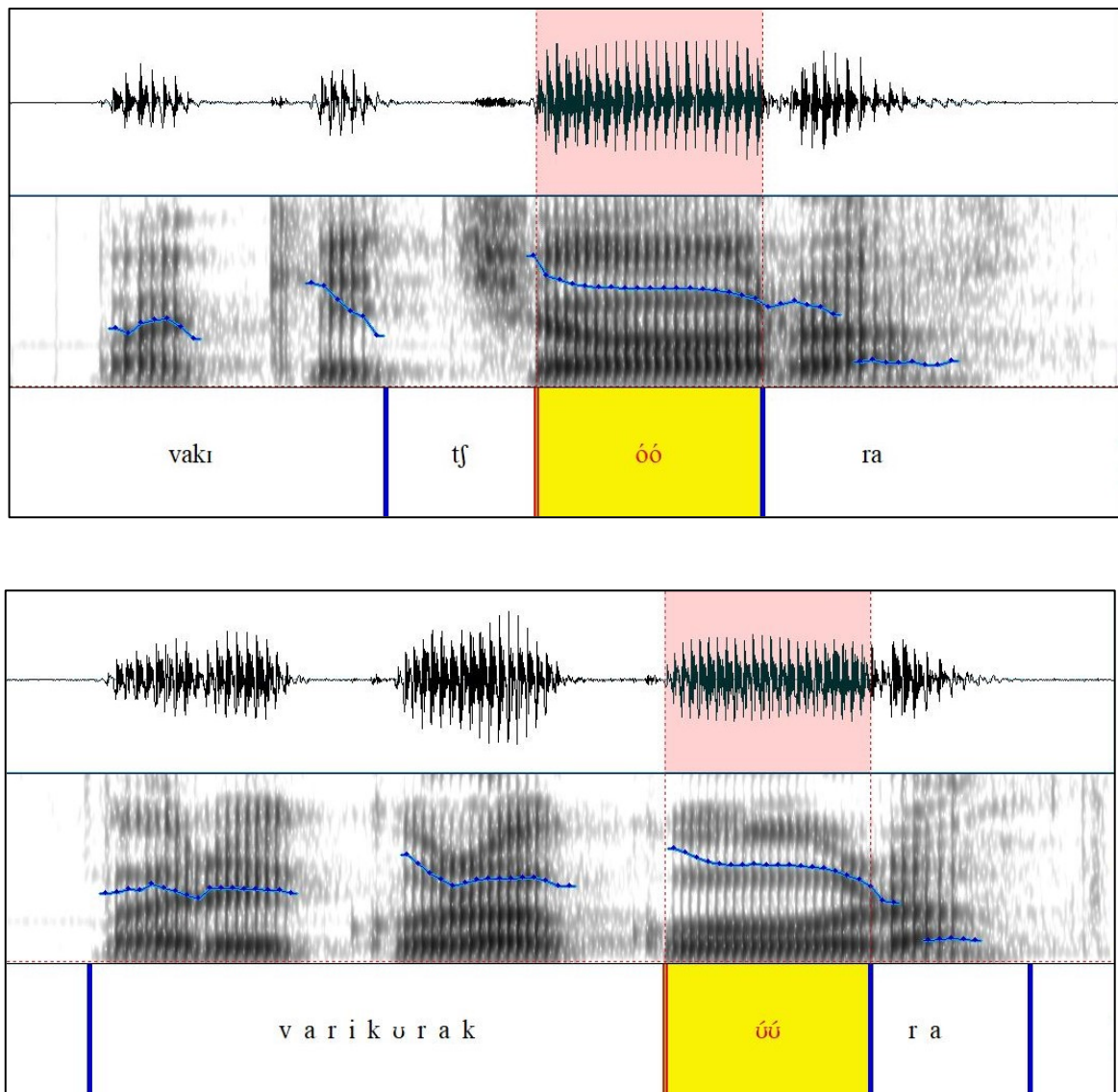
Expected falling tones may be realized as an apparent level tone, thus ‘to cook’ may be realized as *kodééka* – 7 of 28 tokens of this infinitive have level H. It is not entirely clear whether there is an optional phonological neutralization of H and F in this context, instead there may be variation in the realization of fall where some tokens more closely resemble level H rather than falling tone. The strongest indication that there is a categorial phonological process is that one speaker (EM) has on occasion offered pronunciation-pairs side-by-side, such as *árákíishí*, *árákíishí* ‘he will grind himself’, noting that both pronunciations are possible (thus, the speaker is aware of the difference – which might be taken to indicate phonological status if speakers are unaware of subphonemic phonetic differences).¹²

In the following tokens, *índaamwiita* ‘before I killed him’ and *índaachéeya* ‘before I swept it’, pitch falls starting with the penultimate vowels, moreover, the extent of fall is comparable – from 112 Hz. to 90 in *índaamwiita*, from 114 Hz to 88 Hz in *índaachéeya*. The subjective impression of these tokens is different, in that *índaamwiita* sounds like it has a level tone and *índaachéeya* sounds like it has a falling tone. The most significant difference between these two tokens is the duration of the fall: in *índaamwiita* the fall takes 83 msc and in *índaachéeya* it takes 178 msc, if we only count the portion of the fall during the vowel, or 240 msc if we include the entire fall. In other words, the phonological fall in *índaamwiita* is physically realized so rapidly that it is hard to identify the first token as an example of falling tone, thus it may be perceived as level. This is a token-specific limit on detecting falling tone, and other instances of penult fall on [íi] have sufficient duration that the falling nature of the tone is evident.



¹² This is not an endorsement of that theory, just a report that such a belief is held in linguistics.

Compare this to a couple of examples of phonologically unambiguous (and invariant) level H, *vakichóóra* ‘they are still drawing’ and *varikorakóóra* ‘they will release us’.

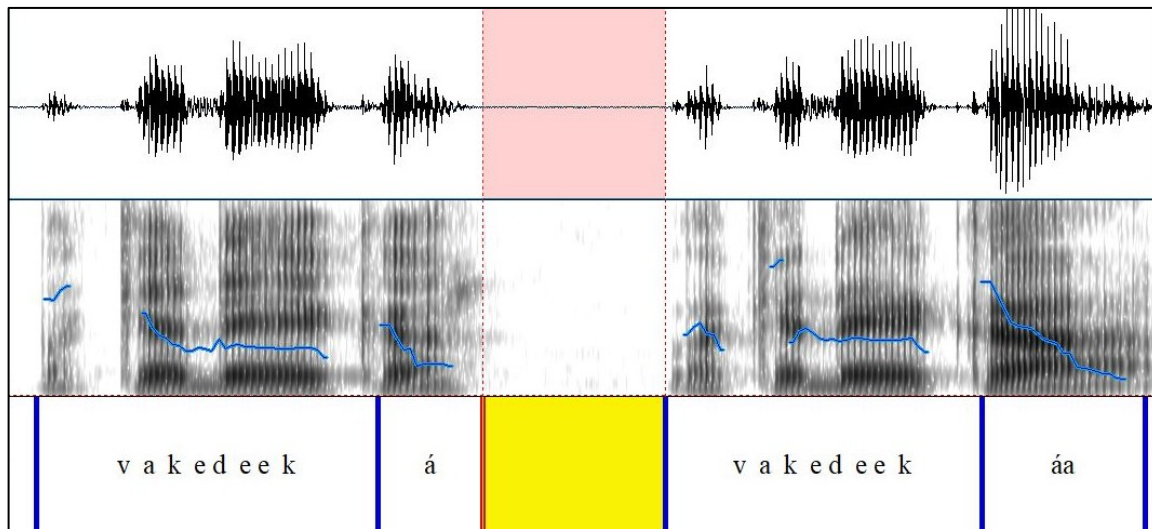


In these examples, there is (if we exclude the substantial raising of F_0 in the initial 15 msc. contributed by the consonant release – a feature lacking in the earlier example *índaachéeya*) much less fall within the syllable (avg. 15 Hz).

A systematic instrumental investigation of the realization of penult fall is necessary to verify this claim, but it appears from informal inspection of a number of tokens that there is substantial variation in the timing of a penultimate fall, with a continuum of phonetic realizations, and not just two outputs which would indicate an optional rule. The practical phonological diagnostic for phonological fall vs. level H is simple: a form which seems to have level H when it ought to have fall will eventually be pronounced with per-

ceptible fall, given multiple tokens, but a real, phonologically-level H will not. The fact that penult fall versus level H are potentially confusable at the phonetic level leads to a prediction of future sound change, and also past sound change. Very briefly, the noun tone systems of Tiriki and Logoori are very similar, but lexical correspondence between level H versus fall on the penult is one area where the systems do not match as well.

In utterance-final syllables, which can be contrastively long in Logoori, there is no contrast between F and H. Level H tokens can occur as well as F tokens e.g. _[em]yɪká'ráá 'he is cutting himself', _[em]vaaníí'mbíráá 'they are singing for me'.¹³ However, there are phonetic reasons to treat final H on long versus short vowels differently: pitch on final [áá] is much higher than it is on final [á]. When a single final H is on a long vs. short vowel, pitch falls in both contexts, but the starting point of the fall in the case of a long vowel is higher than in the case of a short vowel. Compare *vakedeeká* and *vakedeekáá* below, two grammatical variants of 'they are still cooking'.



In position before the penultimate syllable, Fall is almost entirely non-existent. The most common context where it appears in transcriptions is when a long vowel with H immediately precedes an independent H, for example *wéendéve* 'chairman', which is also realized *wéé'éndéve* and *wéé'ndéve*. The realization of phonologically concatenated H tones is discussed below, but it is a general rule that when adjacent syllables have their own H tones, the second tone is realized in a lower register, notated with the downstep marker (*wéé'ndéve*). The timing of the register drop is somewhat fluid, and *c'v'vc'v* may be realized as *c'v'vc'v* or *c'vc'v*. Such differences were noted in transcriptions since it was not certain that there was no contrast, but phonologically speaking, *c'v'vc'v* or *c'vc'v* appear to be translatable into phonologically uniform (surface) *c'v'vc'v*. Underlyingly, though, this surface form may derive from *c'v+vc'v* or from *c'v'vc'v*.

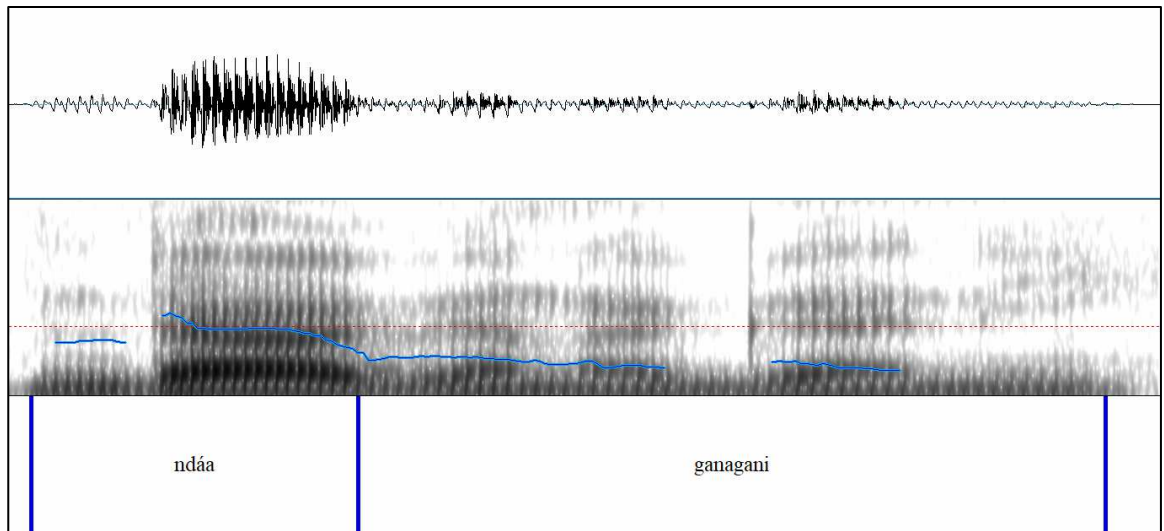
Another context where phonological F is widely encountered is in certain verb inflections where a grammaticalized property of 'prominence' is given to the subject pre-

¹³ Phonetic fall is much more common compared to level H in final position.

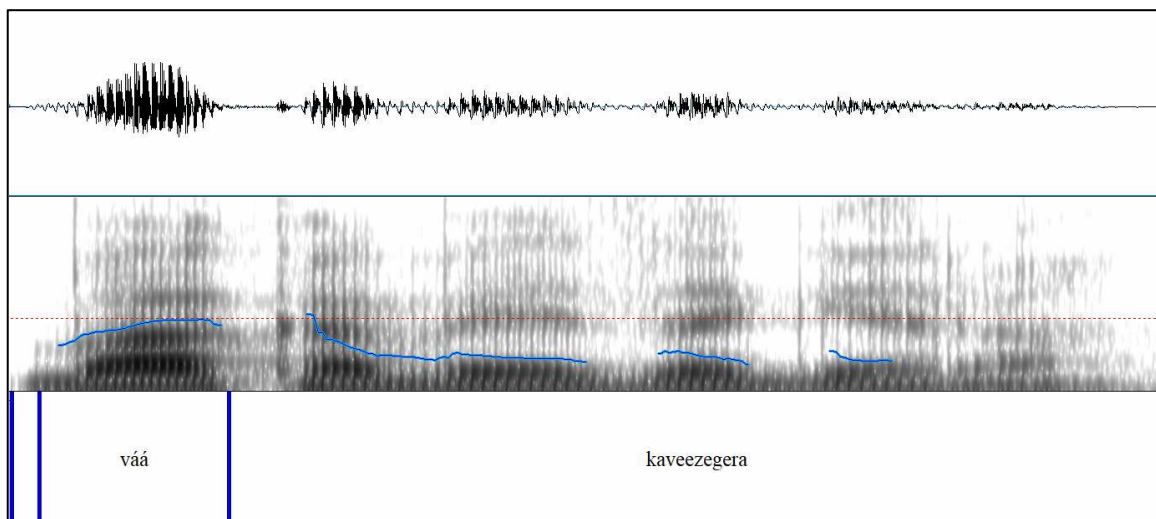
fix (related to focusing on the completion of the action). This prominence is realized as vowel lengthening in some contexts but as a H tone – phonetically as F – when the subject prefix stands before a vowel, as in the following examples.

ndáaganagani	‘I have now thought’
yáakakuza	‘he has now died’
váakaveezegera	‘they have now belched’

We can see from the pitch track of *ndáaganagani* that there is a drop within the first syllable (from 132Hz to 104Hz), but in the center of the syllable, pitch is flat. The auditory effect is that there is a slight fall.



Compare the above token with [váákaveezegera], where pitch does not fall, instead it rises by 10Hz.



The difference in tokens is either purely phonetic implementation, or the result of a late optional phonological rule where H alternates with Fall. The pattern will be treated as phonological, since there is a noticeable similarity between this pre-penult falling pitch and the clearly-contrastive fall found on phrasal penults. Such a Fall also arises at the phrasal level from the merger of a final H toned vowel plus an initial L toned vowel.

mbw-ímdara	‘1 dog’
mb-ééneengero	‘give me a brew pot’
naahíim-íngorovi	‘he will hunt the pig’
rw-áakaraangizi	‘when he fried up’
rw-áacheerizi	‘when he greeted’
n-íngorovi	‘with a pig’

The combination CV́+V may also be realized with level H, cf. *n-íínama* ‘with meat’, *n-éérefu* ‘with 1000’.

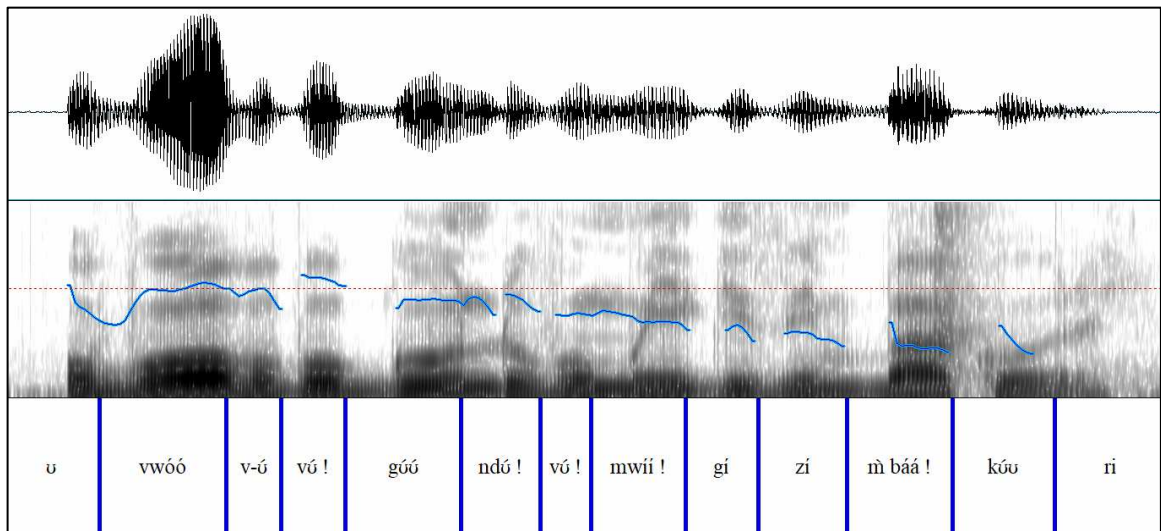
The main tone-inventory issue, then, is what the status of falling tone is. It is almost, but not quite, a positional variant of (level) H on a long syllable. The difference does enter into grammatical and lexical contrasts, therefore it must be included in transcriptions. The difference is often neutralized, and it is often phonetically uncertain whether, in a given token whether Fall has optionally (phonologically) become level H. There is generally no doubt as to what the deeper phonology of the language has (there are level Hs and falling Hs), but the phonological output for specific tokens can be unclear.

3.2.2. INTERSYLLABIC PHONETIC EFFECTS

The phonetic realization of tone as a function of neighboring syllables is complex, to the point that it can be difficult to say which properties are phonological rather than phonetic (which implies decisions about transcription), and it is challenging to say how the abstract phonological representations of the language relate to phonetic realizations. In terms of understanding how phonological tone translates into physical pitch, the primary challenge is sorting the intersecting factors into a system of phonetic principles. Ideally, a H tone would be realized as a relatively high pitch and L is a relatively low pitch, therefore if you know the approximate pitch range of H vs L in a language for a given speaker, you may be able to efficiently switch between cognitive tonal categories and physical traces. A level tone ideally has a consistent pitch during its realization but a falling tone has a consistently decreasing pitch. The situation in Logoori is much more complicated, because there are many contextual micro-adjustments of pitch – rules of phonetic implementation. The challenge of tonal phonetics in Logoori is identifying those adjustments, and determining exactly which ones are phonological versus phonetic.

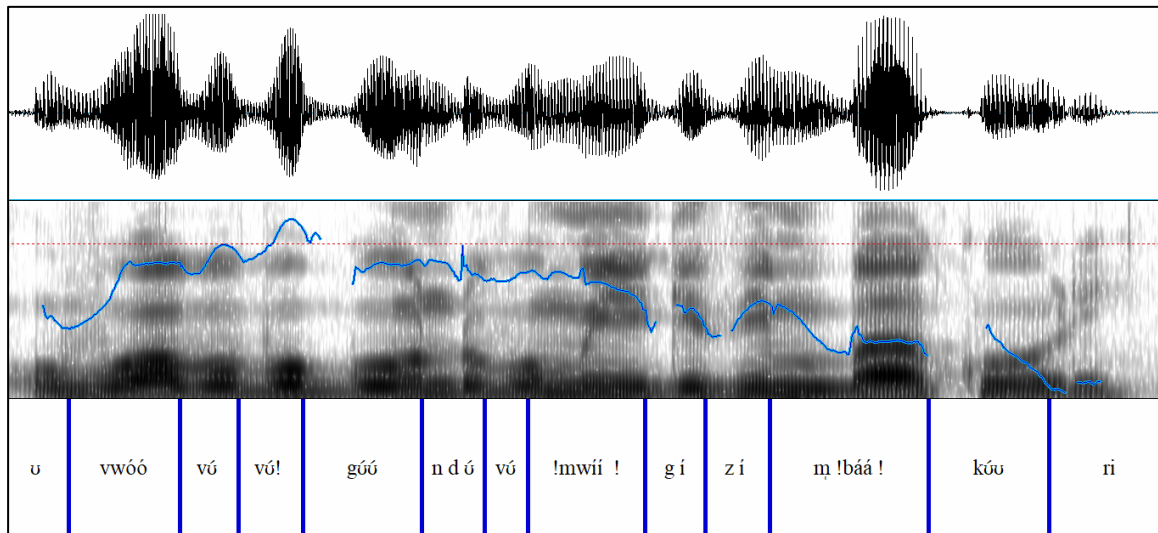
An example of the problem of interpreting physical pitch is seen in two recordings of the utterance [ʊvwoov-ʊvʊgʊʊndʊ v-ʊmwiigizi m’baakʊʊri] ‘the teacher’s rotten mushroom in the bowl’ (tone omitted), composed of the toneless nouns *ʊvwoova*,

the adjective *ovógovundó*,¹⁴ the associative connector /vú-a/, the noun *ov+mwíígizi* where the prefix *ov-* bears an underlying H tone that deletes in certain phrasal contexts, and the locative-marked form /mó+báá'kóri/. Ultimately, we want to know the systematic phonological representation of this combination of morphemes, but we only have direct access to physical pronunciations. The challenge is understanding how to arrive at the correct surface phonological form, based on such actual pronunciations. Simply combining the individual words, with suitable segmental modifications (reduction of vowel sequences, reduction of /mó/, deletion of *w* before *ov*) would give us the theoretical form [ovwoov-ovógovundó vómwíígizi m'báá'kóri]. Compare that transcription to the following two pitch-marked spectrograms, the first transcribed as [ovwóón-óvó'góóndó 'v-ó'mwíí'gízi m'baá'kóri].



The second token (gathered on an earlier day) was transcribed as [ovwóón-óvó'góóndó v-ó'mwíí'gízi m'baá'kóri].

¹⁴ This is the underlying tone. On the surface, the final H spreads to the left giving [ovó'góóndó] – except that the question of leftward spreading and appearance of downstep is a hypothesis that needs to first be established, and is the point under discussion here.



There are differences between the two utterances, as well as an overall similarity. The initial syllable [u] in both instances has relatively low pitch, followed by two syllables [vwóóvó₁] with an intermediate pitch, then the highest pitch is found on [vó₂] (the adjective agreement prefix of /ovó¹góóndó/). In the second token, vó₂ is much higher than vó₁. Such differences are familiar in tone languages, and we would not say that this reflects a phonological difference in tones found in one utterance but not the other, such as H versus Extra-High (a difference that exists in Kamba). It might be a result of an optional phonetic rule boosting the pitch of certain Hs, or it may be an example of tone being randomly instantiated at different points in the continuum of pitch realization.

Pitch drops after [vó₂] and remains relatively flat for [góóndó]. The tokens differ, in that [ndó] is noticeably higher than [vó₃] (“of”) in the first token, and the tokens are accordingly transcribed differently. A phonological treatment would be that downstep is present after [ndó] in the first token, and not in the second. But it is possible (indeed likely) that the lack of apparent downstep here is just an example of the lower end of pitch-drop that characterizes downstep in this language, that is, the degree of drop is very small. Pitch then falls over the syllable [mwíí] – to a greater extent in the second token. In light of other pitch facts discussed in this section, it is worth noting that the pitch of preceding [vó] does not rise significantly in the second token.

In general, the pitch excursions in the second token are larger than in the first. Otherwise, the overall pitch pattern in the tokens is essentially the same, except the lack of downstep in ndó_vó in the second token, a divergence that we will for the moment dismiss as a mystery. What is primarily of interest is figuring out how to relate the underlying form /ovwoov-ovóguundó vómwíígizi m’báá’kóuri/ to the surface transcription [ovwóóv-óvó¹góóndó (’)v-ó¹mwíí¹gízí mbáá¹kóuri]. Coming to terms with Logoori tonal phonetics requires not just knowing what the underlying contrasts are, it also requires understanding the adjustments that are potentially but not necessarily part of the phonological form. That is, one has to understand the phonetic rules of tone realization in the language.

There are three general trends at work in relating the abstract tonal form to actual pronunciations as embodied in the surface transcriptions, which are perhaps phonological

or perhaps phonetic. A first general property of the language is that H tone spreads to the left, explaining why the underlyingly toneless vowels at the left edge of the utterance have higher pitch. This is arguably but not self-evidently the result of a phonological spreading rule, Leftward Spreading. Second, whenever two H tones come together, pitch register is always lowered – they are separated by a downstep (though in a few tokens, the lowering is slight). Since surface HH and H¹H are unpredictable, some contrastive property but be present – transcriptionally, that is the downstep marker. These two facts (spreading and downstep) are related, in that the automatic lowering which constitutes downstep does not apply *within* the span of a single H which undergoes (presumed) Leftward Spreading, but otherwise is always present between any two adjacent H toned syllables. Segmentally-oriented transcription methods have the defect that the distinction between a single H with a multi-syllabic domain cannot be easily written distinctly from a sequence of Hs over a similar domain. The downstep marker is a surrogate for marking the phonological difference between single H on two syllables vs. separate Hs on adjacent syllables.

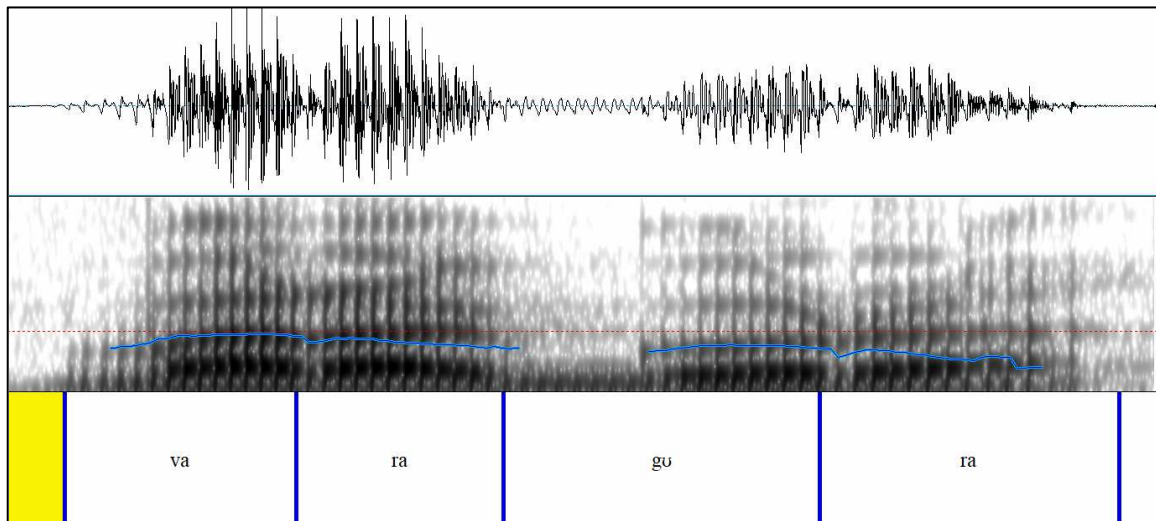
A third phonetic process is seen in both tokens, that the last H tone in a sequence of Hs has raised pitch, but again the status of this process depends on how these patterns are analyzed. The pattern might reflect a local process of “pre-L boost”, or it could reflect a tendency that the physical peak of a H (associated to multiple syllables) is at the right of the H’s domain. The discussion in this subsection starts with the possibility that these are all phonetic processes and therefore the phonological form of the utterance is [ʊvwoov-ʊvóguundó vómwígizi m’báákóori]. We will ultimately (but not overwhelmingly) conclude that Leftward Spreading is phonological, which then simplifies, indeed trivializes, the account of downstep as well as the raising of pitch before a downstep (or L).

Leftward Spread

An underlyingly H will typically spread leftwards to preceding vowels, which is phonologically treated and notated in all previous tone-marked works. For example, *kwaakogora* ‘we bought’ has no H tone and *ekékóómbe* ‘cup’ has penultimate H. In the phrase *kwáákógór-ékékóómbe* ‘we bought a cup’, the vowels preceding the noun (may) have a higher pitch. In the citation form of the noun, there is generally raising at least of the syllable *ke*, thus *ekékóómbe*. The essential phonological question is whether there is a categorial phonological rule spreading H to the left – a phonological form [kwáákógór-ékékóómbe] implying a tone spreading rule – or is the phonological form [kwaakogór-ekékóómbe], and is the appearance of higher pitch due to phonetic implementation?

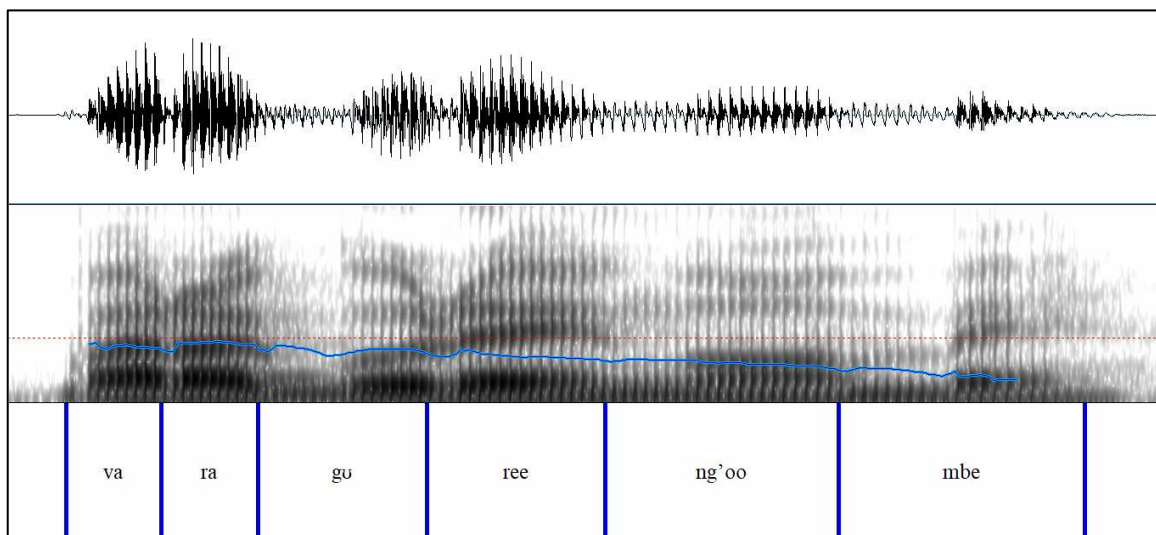
In order to answer the question, we must first look at the difference between an all-L utterance and an utterance with a penultimate H plus a number of preceding toneless syllables. The first kind of utterance is relevant in setting an expectation for what L looks like, and how H might look different from L. Consider an example of the utterance [varagura] ‘they will buy’.¹⁵

¹⁵ This set of examples is drawn from a single session where the speaker (EM) maintained the same general pitch range and was focused on producing “canonical pitch patterns”.



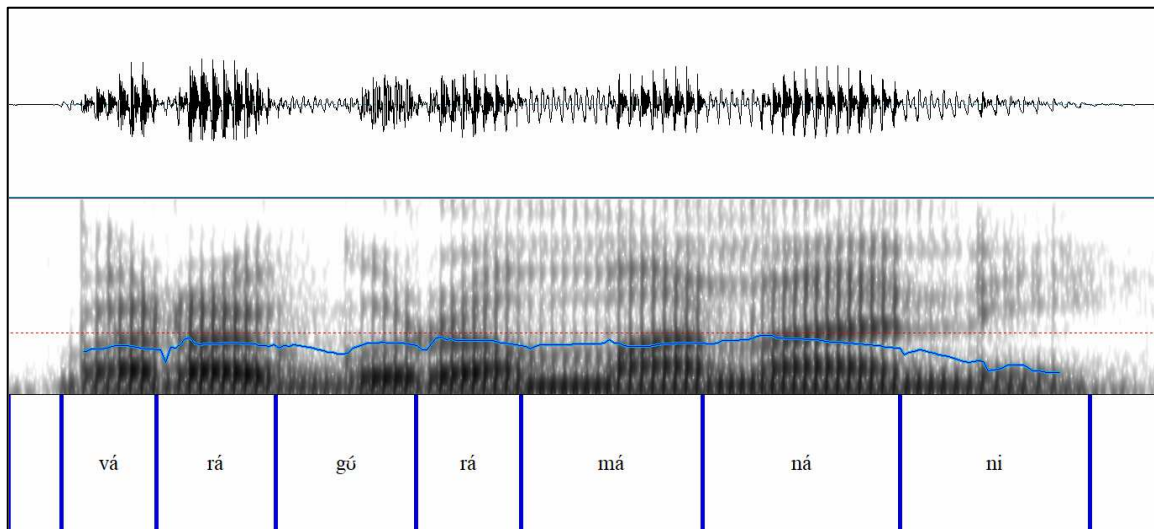
In an utterance with no H tones, pitch gradually declines throughout the utterance. In [varagura], pitch declines from 113Hz to 102Hz in the span that excludes the final syllable (final syllables are expected to have special pitch properties, and pitch is often not computable in a prepausal syllable).

In [varagor-eng'oombe] 'they will buy a cow', we find a similar decline of pitch from 114Hz to 93Hz over the non-final syllables.



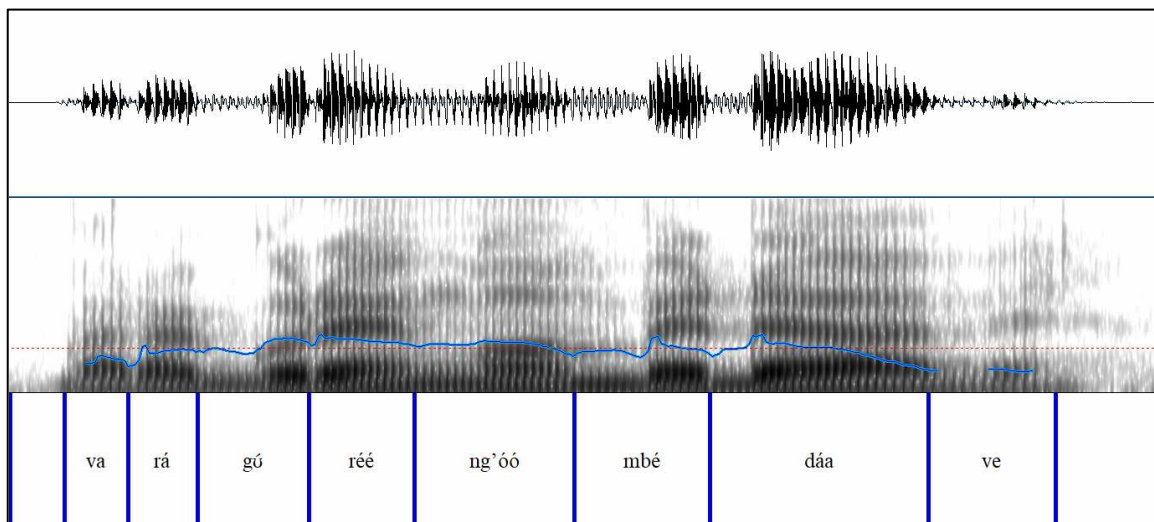
This raised-to-lower pitch decline is a hallmark of phonologically L-toned final spans.

Now compare the preceding two examples to [várágórá mánáni] 'they will buy monsters', with the noun /amanáni/ which has penultimate H. Rather than declining, initial pitch is relatively flat, and actually rises a bit at the penult from 105Hz to 108Hz.



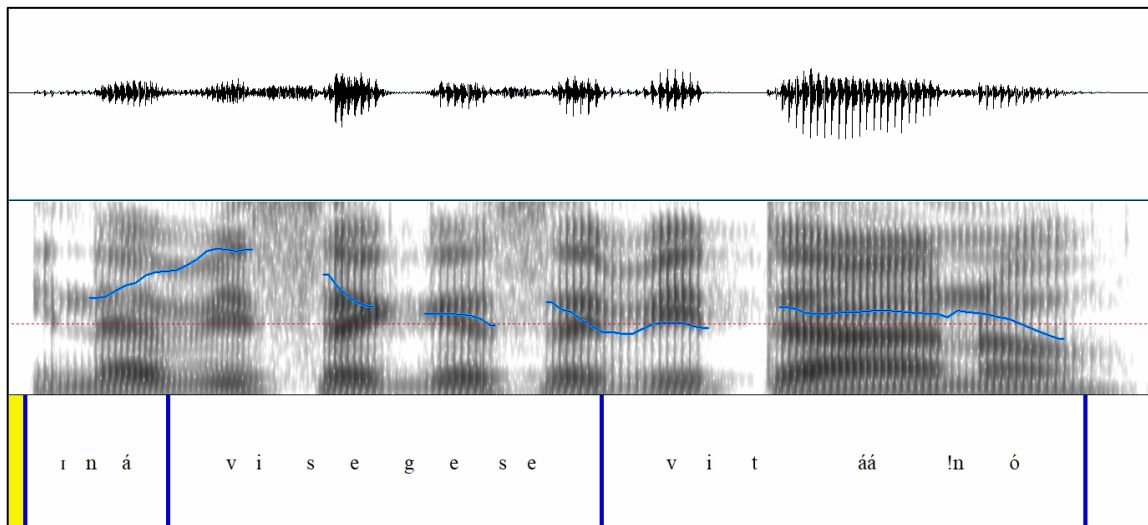
While we can easily characterize the difference between L* versus H*L from L*HL as a degree of anticipatory raising in L* (a raising which enables continuous pitch declination) versus a relatively flat pitch pattern in H*L (presuming a rule spreading H tone), the perceptual challenge is that the initial “H” tone of [várágóra mánáni] is physically lower than the initial “L” tone of [varagor-eng’oombe]. The all-L utterance confusingly starts at a higher pitch than the H*-initial utterance. This “continuous drop” strategy is how all-H and all-L contrasts can be identified: absolute pitch is not particularly relevant.

A related puzzle is the difference between [várágóra mánáni] and [várágór-ééng’óómbé dáave] ‘they will not buy a cow’, from /varagora eng’oombe dáave/. In the former example, the pitch of the first syllable of the utterance is close to that of the second syllable (102Hz vs. 105Hz), whereas in [várágór-ééng’óómbé dáave], there is a greater separation of pitch in the first two syllables (90Hz vs. 96Hz).



This 3Hz difference in the first syllable is audible and might be categorical (maybe interpolation does not start with the initial syllable in [várágór-ééng'óómbé dáave] but does include the initial syllable in [várágór-ééng'óómbé dáave]). Or, the difference may reflect a difference in specific realization value that arises at random from token to token. A phonologized difference [várágór-ééng'óómbé dáave] vs. [várágór-ééng'óómbé dáave] does reflect a recurring distinction in realizations, in my experience, but that may simply reflect a categorization strategy of this hearer. This is a matter that requires extensive experimental testing.

The main reason for treating Leftward Spreading as phonological is that it is impossible to say when H is anticipated, based solely on phonetic criteria. For example, there is systematically no such spreading in the syllables before the penult in *máviségese vitáá'nó* 'towards (the) 5 roof peaks', which exemplifies the *ma-* construction. The citation noun is *viségese*, which becomes *máviségese* by prefixation of the directional prefix. A characteristic of this construction is that the noun following *ma-* is L toned, meaning it loses lexical H tones, and is not assigned H by any phonological process. Observe that while final H spreads to the penult in *vitáá'nó* (splitting of the final H into H'¹H being a common variant of prepausal H), it does not spread further left. We see that pitch is higher over the last two syllables and lower in the intervening span (allowing for general rightward shift of the pitch peak by up to a syllable, and the one-syllable transition to the non-H target pitch).

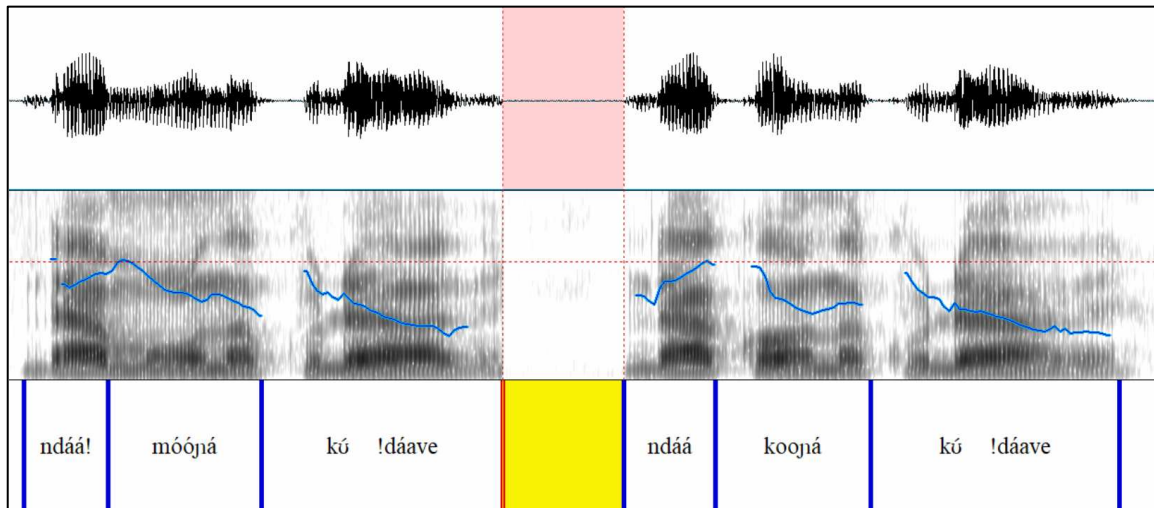


A phonological account of this difference in pitch realization which avoids phonological tone-spreading is to posit a surface contrast between marked L and unmarked Ø, thus *máviségese vitáá'nó*. Grave accent effectively indicates where would-be Leftward Spreading is blocked. There is no phonetic difference between marked L and a tonally unmarked vowel – the distinction would be based strictly on this particular phonological behavior.¹⁶ While this specified-L treatment may avoid a rule of Leftward Spreading, it

¹⁶ Note though that grave (and acute) accent *are* used (somewhat unsystematically) in this work to mark syllabicity of nasals, rather than introducing the syllabicity diacritic in [ṁ, ṇ]. Certain traditional ortho-

necessitates a rule of L insertion. At this level of analysis – questions of transcription – this is not a valid basis for adopting a more opaque transcription, even if it turns out to be a correct phonological analysis of the data more narrowly transcribed.

Below we see an example of contrasting tone patterns, where H spreads from the enclitic *kó* into the preceding verb. There is a difference between *ndáá' móóná kó 'dáave* ‘I’ve never gossiped’ and *ndáákoóná kó 'dáave* ‘I’ve never helped’, brought about because /moóna/ is a toneless (L) verb stem but /kóóna/ has lexical H, which, while phonologically deleted still has a blocking effect on leftward spread of H. In *ndáákoóná*, pitch rises after the initial fall into the L syllable, whereas in *ndáá' móóná* there is a continuous fall from the higher-register H to the lower-register H.



Again, these examples could also be transcribed as *ndáámoóná kó 'dáave* versus *ndáákoóná kó 'dáave*. In eliminating Leftward Spreading from transcriptions and substituting grave accent to indicate the leftward limit of a phonetic analog of Leftward Spreading, we could also eliminate downstep markers in surface transcriptions, instead understanding that any two H tone marks are always separated phonetically by downstep (otherwise, two adjacent H tone marks always reflect application of Leftward Spreading). But for reasons of clarity, downsteps will not be eliminated and the effect of Leftward Spreading will not be erased. The burden is on the reader to remember that some of these properties may result from applying rules.

A further pair of sentences can be given to demonstrate the phonological nature of Leftward Spreading. In the hodiernal past, we find that lexically H verb roots have no H in the citation form, and L verbs have H on the first two stem vowels. When followed by a modifier with a H – *gáráha* in the following examples – H spreads into the verb in the

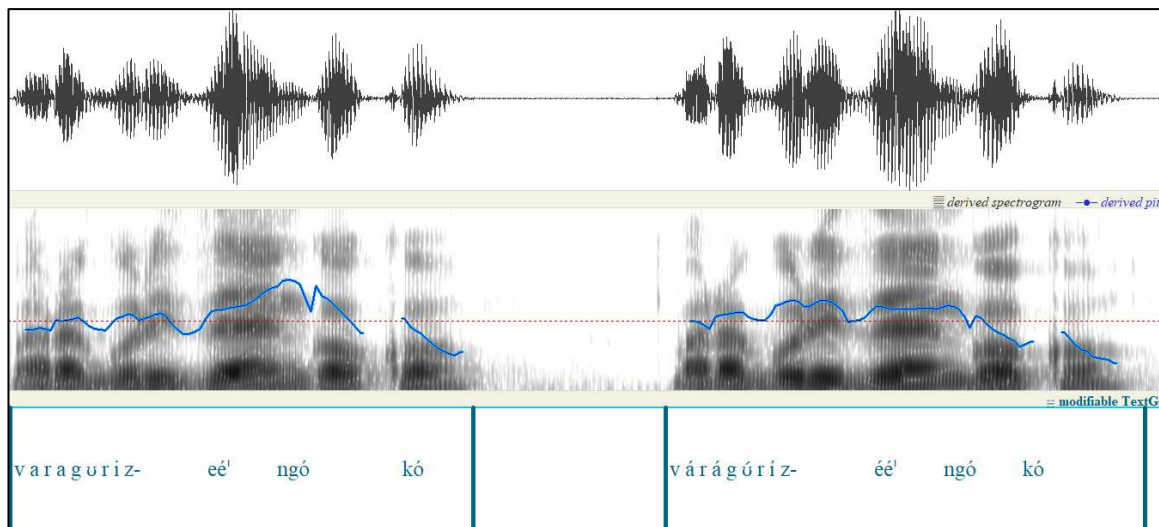
graphic practices namely the use of apostrophe to mark an elided vowel between consonants further eliminates the need for that use of tone marks, since syllabic [m,n] always derive from deletion of a vowel after the nasal, and consonants brought together by vowel deletion can be notated with an apostrophe, thus [óm'mósi] rather than [óm'mósi]. I have since adopted this writing convention, but the majority of my data were previously notated with this distinctive use of grave accent as a substitute for IPA [m̩].

case of lexically L verbs, but not at all in a H verb; and note that the limit of spreading is tense-specific.

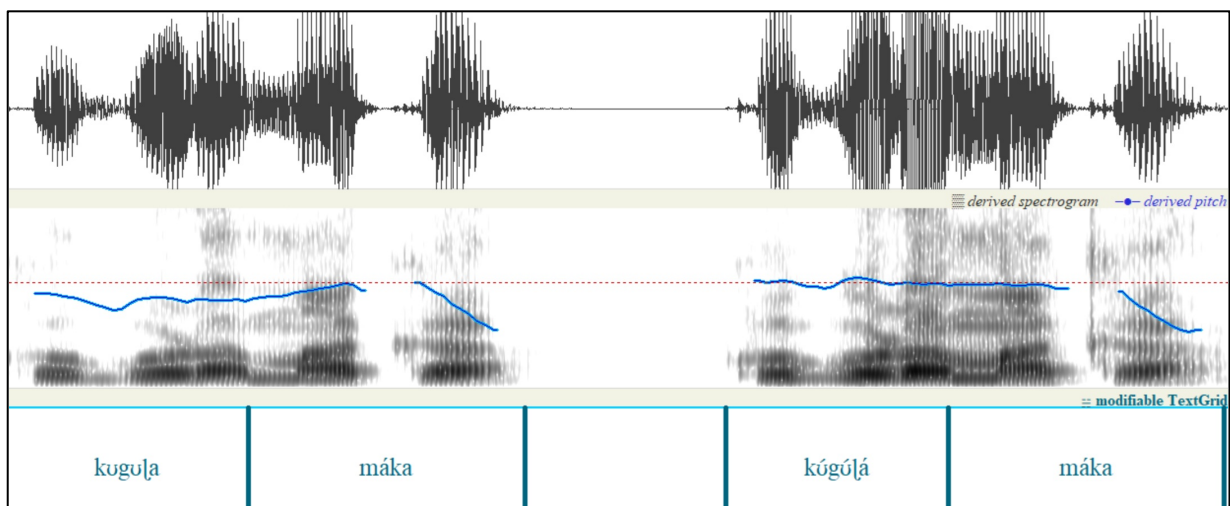
kuvoruganya	‘to stir’
kugírong’ana	‘to turn around’
kóvórógányá gáráha	‘to stir slowly’
kugí’róng’ána gáráha	‘to turn around slowly’
mbóróganyi	‘I stirred’
ngírong’ani	‘I turned around’
mboróg’ányí gáráha	‘I stirred slowly’
ngírong’ani gáráha	‘I turned around slowly’

We can account for failure of LS by positing a distinction /kuvoruganya, kugírong’ana, mbóróganyi, ngírong’ani/, allowing us to deny that Leftward Spreading is a phonological rule, instead relying on a distinction between toneless vowel (allowing Leftward Spreading) and L toned vowels (which block tone spreading).

There is subtle but fairly compelling evidence in Logoori for distinguishing phonologically L vowels from phonologically toneless vowels (Ø-specified vowels), which comes in the form of blockage of Leftward Spreading in specific contexts. There are at least two segmentally identical verb conjugations in Logoori, where one form not only has no H tones for any verb, the form blocks Leftward Spreading; but the other form has a logically-governed pattern where lexically L verbs are again all-L *but* Leftward spreading can apply into the verb. The two pairs of forms are (a) the immediate future versus the counterfactual, both having the segmental form SP-ra-STEM, and (b) the “general recommendation” subjunctive with 1pl subject versus the infinitive. To take an example of (a), in conventional spelling *valaguliza engoko* which is pronounced either as [várágóriz-éé’ngókó] meaning ‘they will sell a chicken’ or as [varagoriz-eé’ngókó] meaning ‘if they sell a chicken’. The difference between the two tonal pronunciations is the result of different tones on the verb *valaguliza*, where the conditional has a final marked L which obliterates any preceding H, whereas the immediate future just has the lexical tones of the root plus any Hs from an OP. The tenses are tonally distinct in some citation forms such as *varákáraanga* ‘they will fry’ vs. *varakaraanga* ‘if they fry’ or *varázígoriza* ‘they will sell them’ vs. *varazigoriza* ‘if they sell them’, but in the case of L verb roots with no OP, the two tenses neutralize in pronunciation – *varagoriza* ‘they will sell’, ‘if they sell’. We can still distinguish the forms phonologically as *varagoriza* ‘they will sell’ and *varagorizà* ‘if they sell’, though in this particular context the two are pronounced the same. When a H toned object follows, a phonetic distinction emerges where marked L tone presents Leftward Spreading and Ø tone does not. The physical difference is seen in the spectrogram below.



The same difference arises in the sentence pair [kugola máka] ‘we should generally buy charcoal’, which has a specified L that blocks tone spreading, and [kúgólá máka] ‘to buy charcoal’, with application of Leftward Spreading.



These differences of pronunciation can be memorialized with a grave accent, vis [varaguriz-eé'ngókó] ‘they will sell a chicken’ versus [varaguriz-èè'ngókó] or [kugolà máka] ‘we should generally buy charcoal’ versus [kúgólá máka] ‘to buy charcoal’. Or, we can record the effect of Leftward Spreading in the transcription.

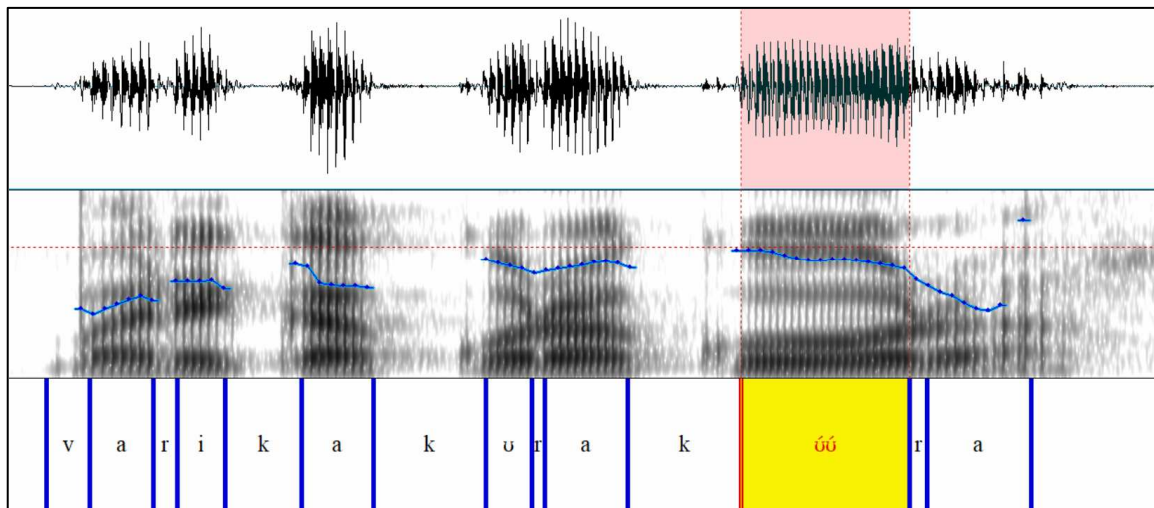
This freedom of analysis means that we could either affirm or deny that Leftward Spreading is part of phonology (alternatively, it is part of phonetic implementation), if we use marked L tones in transcriptions to signal categorial limits on that phonetic process. This approach is not be adopted here – for transcriptions – for the simple reason that the rules for interpreting transcriptions become complex to the point of making misleading

suggestions about pronunciation. As it happens, there are actual phonological arguments for positing a specified L as the triggering factor, but it is horse-cart inversion to represent pronunciation based on a supposed analysis of pronunciation itself. The phonetic facts are clear, as seen above, we will consider the theoretical implication of those observed facts later.

Pitch interpolation

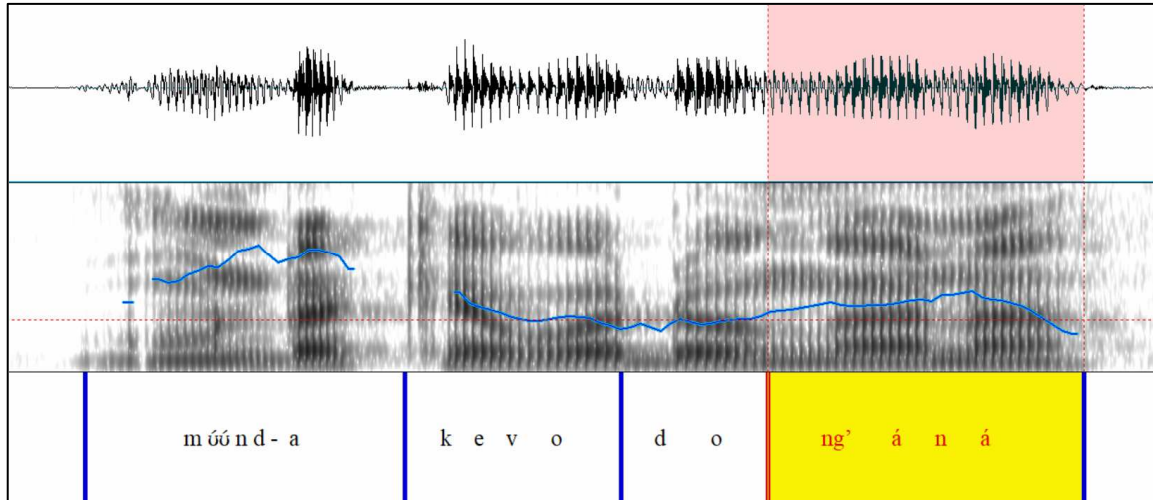
A basic challenge in tonal parsing for Logoori is deciding what phonological tone pattern to attribute to a sequence of toneless syllables. How does one distinguish $\emptyset\emptyset\ldots\emptyset\emptyset\#$ from $\emptyset\emptyset\ldots H\emptyset\#$ or $\emptyset\emptyset\ldots\emptyset H\#$? You might expect that the procedure is simply to detect whether and where pitch goes up, but there are long-distance pitch interpolation processes which make tone-identification difficult.

One complicating factor is that H does not always spread to the left (even setting aside absolute blocking as in *mavítabù víra*). One can find a gradual interpolative rise in pitch in the pre-penultimate syllables of [varikakorakóóra] ‘they would release you’.



Each syllable up to the penult has a slightly higher pitch than its predecessor, a pattern often taken as evidence that such syllables have no phonological tone, and instead receive their F_0 value by an phonetic interpolation. Various phonological interpretations of this particular token are possible – [varikakorakóóra], [varikakorákóóra], [varikakorákóóra] or [varikákórakóóra]. Reasonable principles of phonology and phonetic interpretation may be posited where H optionally spreads phonologically to the left up to some point, but the upward pitch trend still takes time within the two or three syllables that constitute the domain of the phonological H. One can hope that a focused phonetic investigation will eventually identify the exact range of variation in pitch realization, but for the time being we will simply say that sometimes there is an apparent categorial raising of pitch as in the previous example [várágórá mánáni], versus the gradual rise in [varikakorakóóra].

In the following example [móúnd-akevodóng'áná] ‘the person who is still going around’, there is a final phonological H, which seems to spread to the preceding two syllables (judging from the pitch pattern), stopping at [vo].¹⁷

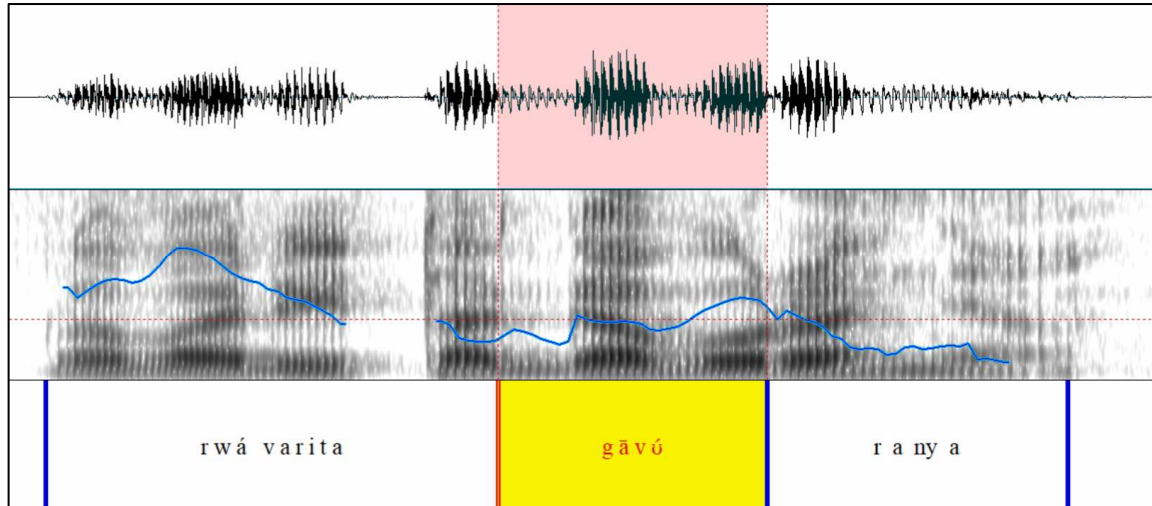


We find more pitch raising in penultimate [ng'a], and some pitch raising in antepenultimate [do] – but no raising in [vo]. We could then notate this limit on leftward pitch interpolation with a grave accent, thus [móúnd-akevòdong'áná]. The underlying verb root is /vódong'an-/ and this verb form is inflected with the M2 tone pattern where a H appears at the end of lexically H verbs, with the preceding lexical tone being deleted. Judging from the evidence of the limit on Leftward Spread, we might instead say that H converts to L, which constrains the possibility of pitch raising on /vó/.

Apart from understanding the disposition of the final 3 syllables, other phonetic processes are at work in the first four syllables. The second syllable *nda* has a relatively high pitch owing to the preceding H syllable. H pitch on [móú] carries over to the following toneless syllable. Subsequently, there is an even lower pitch on [ke] followed by the lowest pitch on [vo]. This example introduces post-H pitch roll-off.

In [rwá varitagávóranya] ‘when they won’t divide up’, the second H syllable [vó] is more noticeably higher in pitch compared to the preceding syllable [gá]. This could be because there is one H over two syllables and pitch only reaches its peak at end of the domain of association. Or, it could be because of a process of pre-L boost – H becomes higher before L. Whatever the cause of this raised pitch, it is rather noticeable, and is phonetically transcribed as a mid tone followed by H tone in the spectrogram.

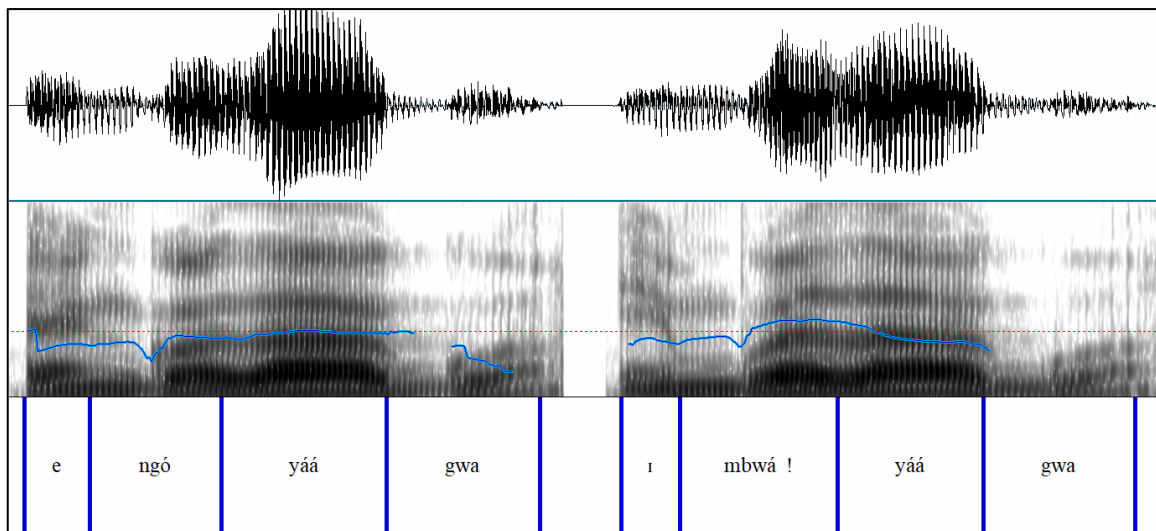
¹⁷ It is a general pattern that a short leftmost syllable in the domain where LS applies has the lowest pitch, a well-known start-up effect in tone languages.



The notation [gā] does not mean that the language has phonological M, it just records a perceptible low-level phonetic property of this particular token, that the pitch of that syllable is intermediate between preceding L-toned [ta] and following H-toned [vó].

Downstep

Logoori also has the contrastive register-lowering operation of downstep, where downstep always and only occurs between two phonologically autonomous H tones. For example, the noun *mbwá* ‘dog’ has a final H, and the verb *yáágwa* ‘it fell’ has initial H. When these two words are combined, a downstep appears – *mbwá* ¹*yáágwa* ‘the dog fell’. We can contrast downstepped H with the HH sequence of *engó yáágwa* ‘the leopard fell’ which comes from /engo yáágwa/ via Leftward Spreading.

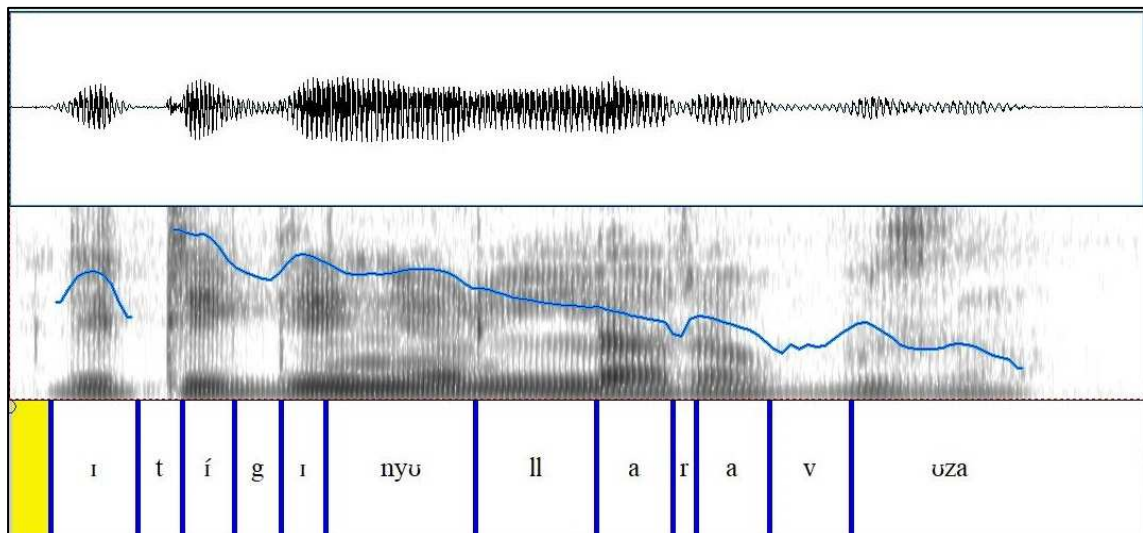


In [ɪmbwá 'yáágwa], the syllable [mbwá] has a peak pitch of 130Hz and [yáá] has a terminal pitch of 110Hz – a drop of 20 Hz. This drop is accomplished partially by lowering the pitch of the following syllable, but also partially by raising the pitch of the first H in the sequence of Hs. Compare also the highest pitch in [engó yáagwa] which is 119Hz, 11 Hz lower than that of the highest pitch in [ɪmbwá 'yáágwa].

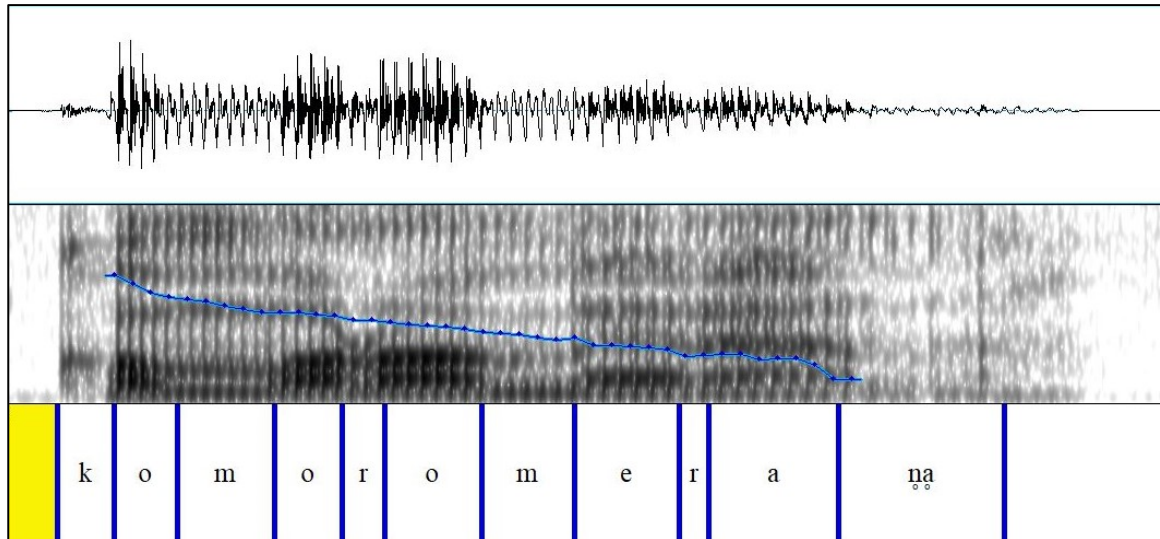
Given a phrasal combination of a word with H followed by a word with H, and underlyingly L toned syllables between, when H tone spreads leftwards, the two H tones abut at the point of the first syllable, and there is downstep, thus *váá'ná varáhi* ‘good children’ from /váána varáhi/. Downsteps can arise within the word when there are two H tones, for example *mí'dógá* ‘cars’, from /mí-dogá/. Word-internal downstep typically involves non-adjacent vowels, and when Hs might be directly concatenated within the word, one of the Hs is generally deleted. Those processes are taken up in the chapters on tonal phonology. As suggested in the previous section, downstep is fully predictable.

Toneless Downtrend

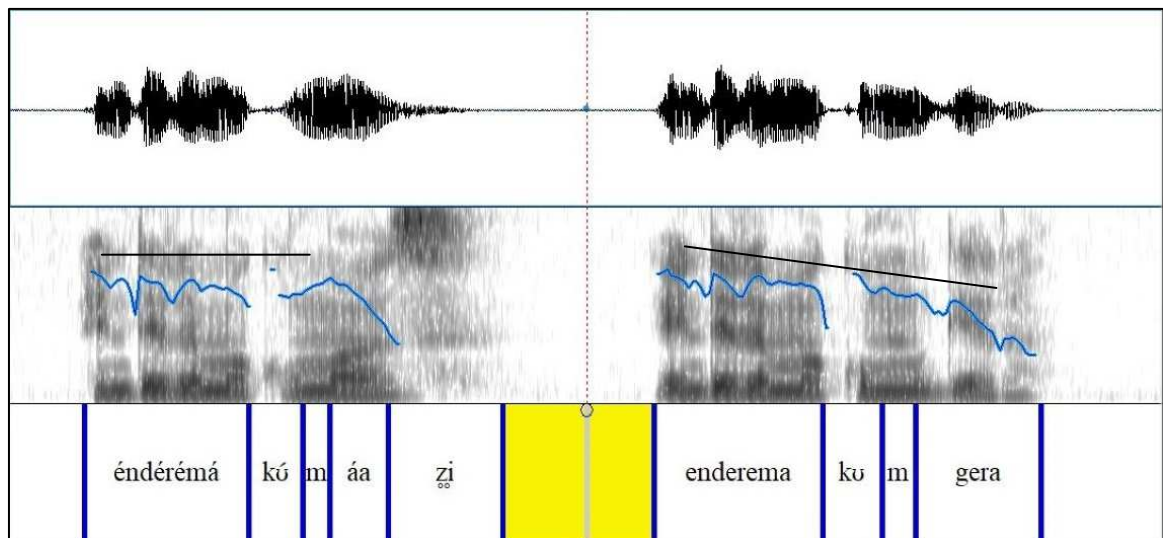
Another noticeable phonetic tendency mentioned above is that pitch gradually descends after H, in a sequence HL*. Phonologically, only the syllable [tí] in [ittíginyo llara voza] ‘just 1 heel’ has H tone, but the pitch trace shows that pitch slopes continuously down from the peak on [tí].



This downward sloping trend does not require a preceding H, as can be seen in the example [komoromerana] ‘to speak for e.o.’.



The general phonetic pattern is that in any sequence of final Ls, pitch falls continuously, meaning that the initial pitch of the first L may be raised compared to later Ls, in order to accommodate this descending pitch profile. This can lead to extreme raising and near-neutralization between L* and HL*. In [enderema kumgera] ‘enderema at the river’, the initial pitch of *enderema* is comparable to that found in [éndérémá kómáazi] ‘enderema at the water’ (these utterances were collected together, in order to control pitch range). The difference lies not in the absolute pitch level of *enderema* in these two contexts, but rather in the fact that in [éndérémá kómáazi], the pitch level is flatter and falls abruptly with the falling toned syllable [máa], whereas in [enderema kumgera] there is a subtle decrease in pitch across all syllables.

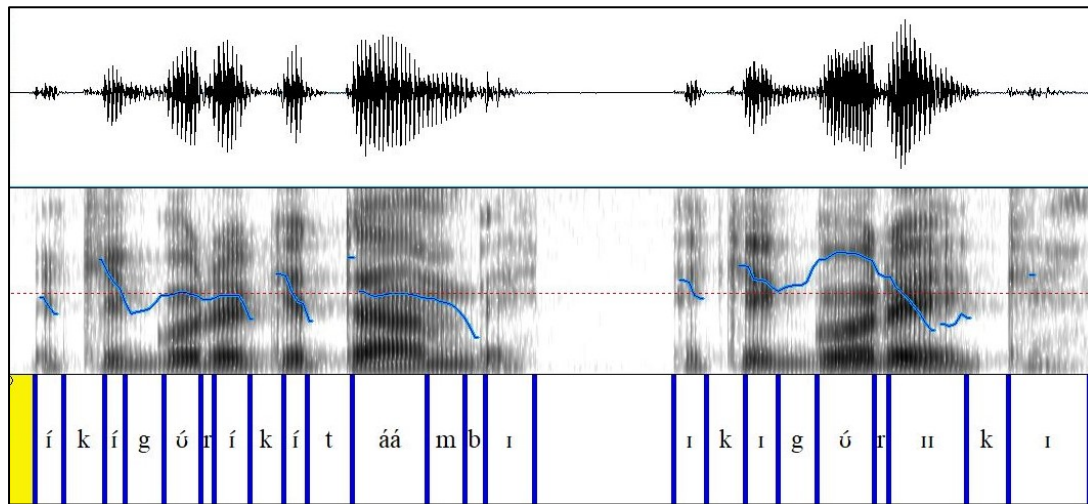


Comparing [éndérémá kómáazi] above, versus earlier [varikakorakóóra] in the discussion of Leftward Spread, and as implied by the different transcriptions, we find gradually increasing pitch in [varikakorakóóra] and abruptly increased pitch in the former. Either

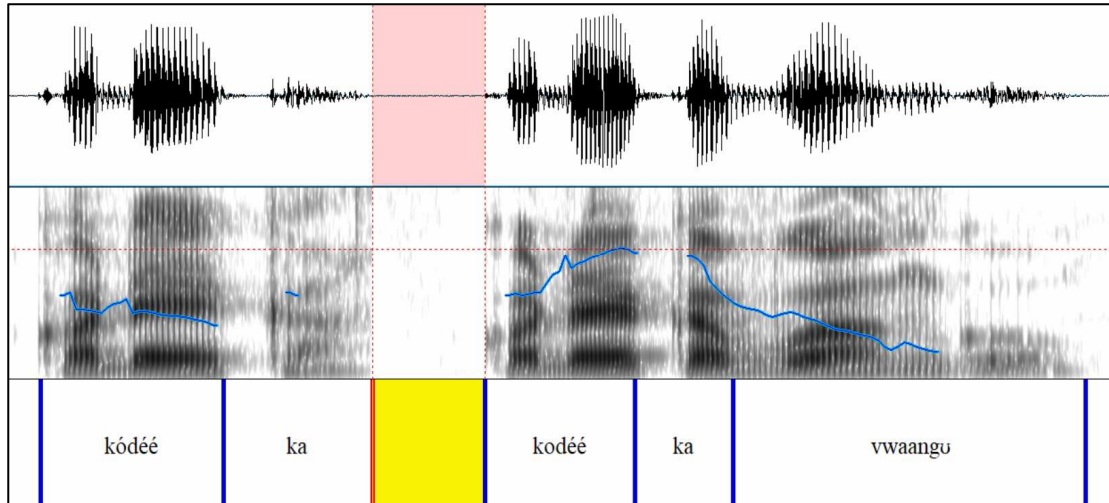
one or both of these phonetic processes is optional, or Leftward Spreading is phonological and is not applied in the case of [varikakorakóúra]. Since there is no clear phonetic rationale behind blockage of Leftward Spread in this case, this is an analytic puzzle, and is one of the facts that underlies the decision to not transcriptionally suppress the effect of Leftward Spreading.

H Boost and Pre-Penult position

A related and often-encountered feature of Logoori tonal phonetics is raising of the pitch of a pre-penultimate H. This can be seen in comparing *íkígór-íkítáámbɪ* ‘tall hill’ versus *íkígór-ɪkɪ* ‘this hill’. The noun *íkígórɔ* is L toned, and in the first example, H spreads leftward from the adjective *kitáámbɪ*. In the second phrase, H is phonologically assigned to the penult of *íkígórɔ* because of the following demonstrative. The examples differ in where the (last) H appears – on the penult versus the antepenult. In the former case with the last H on the penult, F_0 remains at around the same level up to the penult (average 106 Hz), which level is indicated by the dashed line. In the latter context, the first two syllables are at roughly the same pitch compared to *íkígór-íkítáámbɪ* then on the syllable *gú*, pitch rises substantially (average 130 Hz).

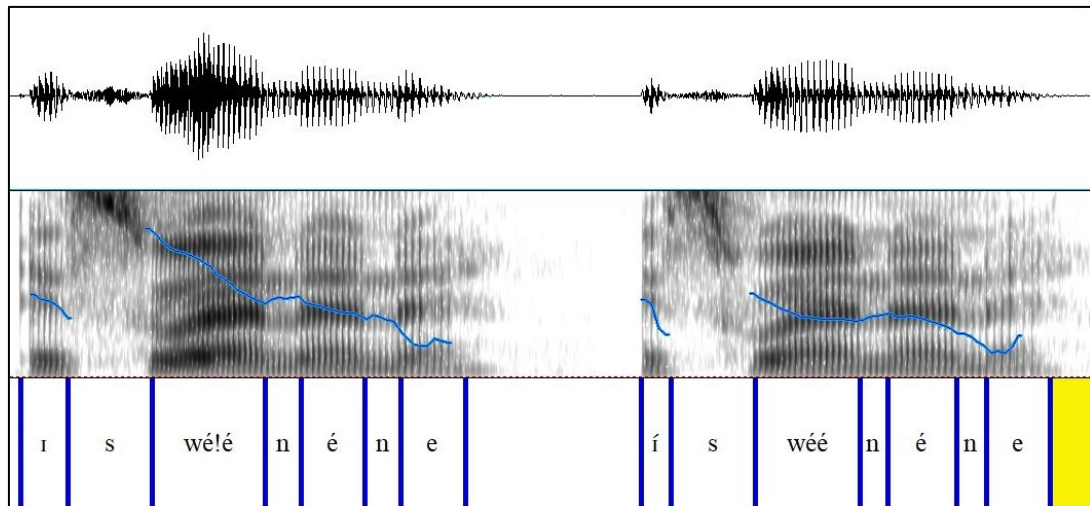


Another example of potential pre-L boost is the difference between [kódééka] ‘to cook’ versus [kodééka vwaangɔ] ‘to cook quickly’, where placing *vwaangɔ* after the verb subjects the H to phonetic pitch-raising (these utterances were elicited next to each other so that a similar pitch range would be maintained).



A further point to notice in connection with pre-H boost is that it only affects the syllable which is immediately before L, a fact which motivates the interpretation of Leftward Spread as a phonetic process rather than phonological category-changing (in which case all syllables should have raised pitch).

Pitch-boosting is seen even when there no overt L tone, in a H¹H sequence. Consider *ɪswé¹énéne* ‘termite sp.’ from /ɪswá enéne/ and *ɪswéénéne* ‘big female chicken’ from /ɪsu enéne/. In the former, the initial H is much higher, and in the latter the H is simply a “copy” of the penult H, which would be lower because it is in the penult.

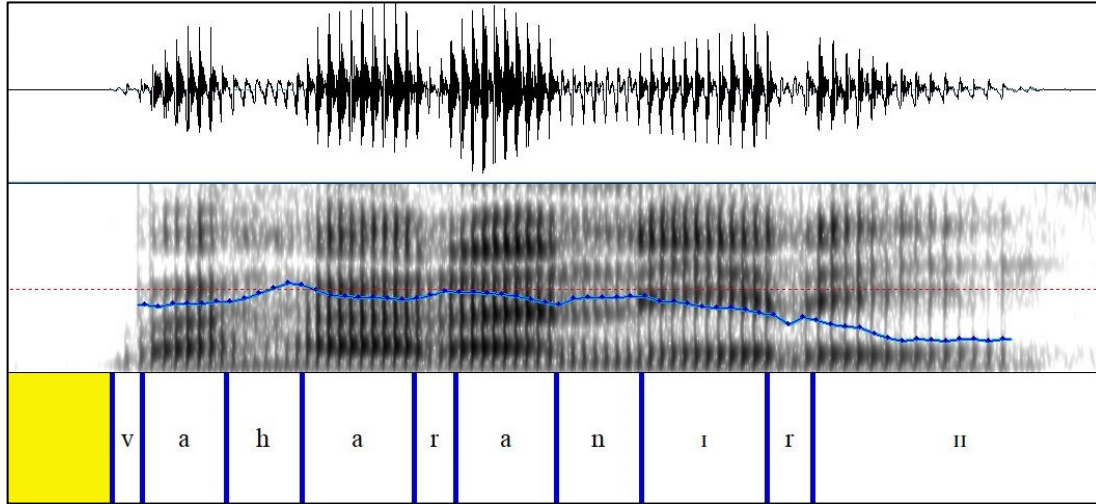


If downstep is a floating L, we can explain the higher pitch on the second syllable by reference to pre-L boost. We can also explain it because there are two autonomous H tones in the former example and only one in the latter, so to make room for two H-defined pitch registers, the first H would naturally have to be raised above the second. Furthermore, the first H in *ɪswé¹énéne* precedes the penult, but the H in *ɪswéénéne* is in the penult. Penult and final H are phonetically lower than pre-penult H.

Penult-Lowering

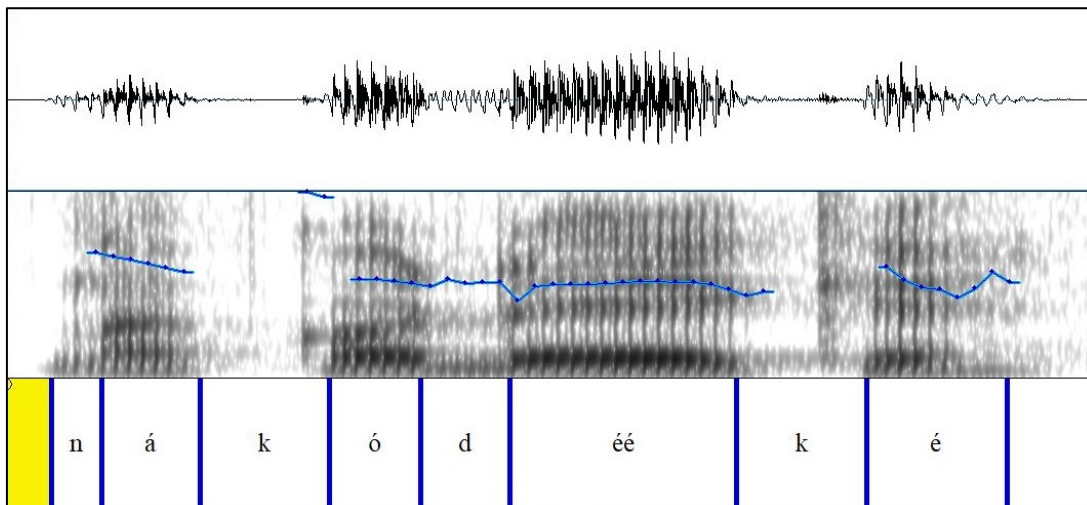
As observed above in [kódééka] versus [kodééka vwaangʊ] a penult H has lower pitch than pre-penult H. Raising in [kodééka vwaangʊ] may be due to pre-L boost, but final L does not trigger raising. Another interpretation is that there is no boost of H, instead there is lowering of a H in the penult – but L is also subject to pitch lowering.

Frequently, F_0 slopes downwards more rapidly in the last two syllables, as illustrated in *vaharanirɪ* ‘they smoothed for e.o’. This may give the impression that the tone pattern is HHLL, though no syllable is phonologically H.



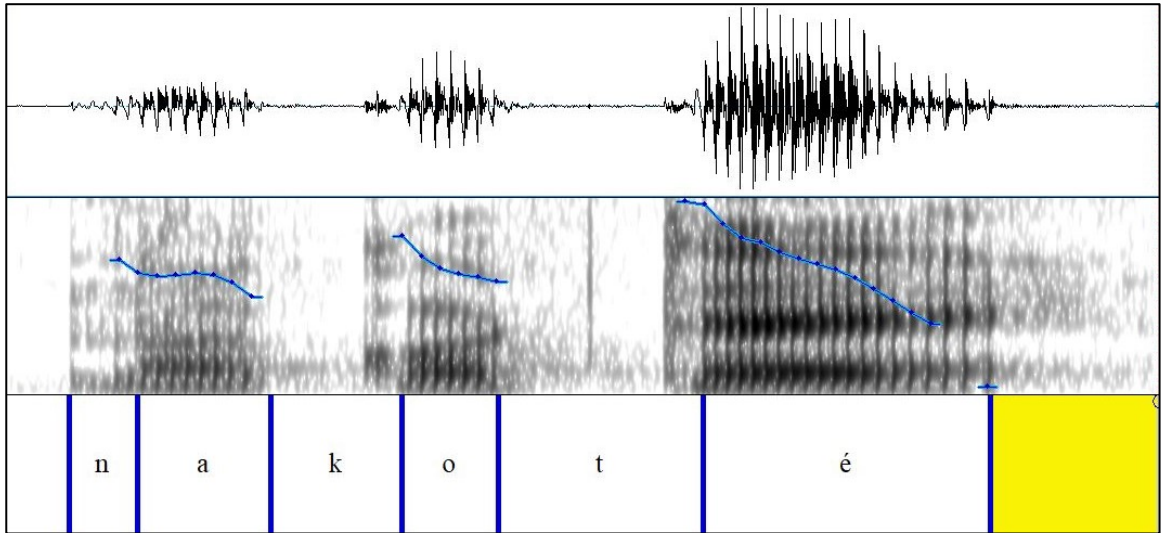
Final Flattening

When an utterance ends L*H, it is often realized as a relatively flat, slightly raised pitch. An example of this is seen in the following token of /na kodeeké/ ‘we will cook’.



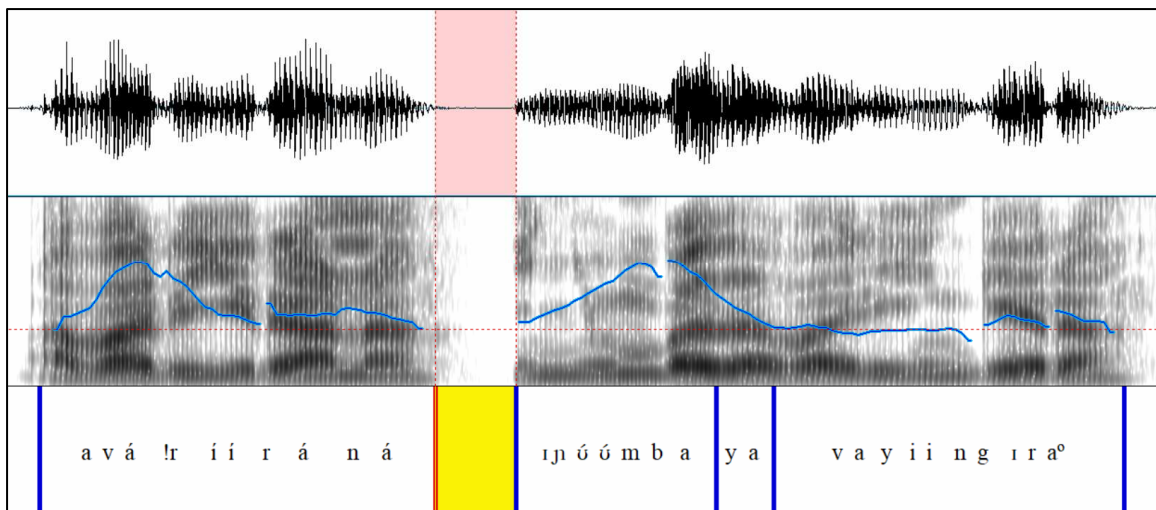
This might be transcribed as [ná kódeéké] (as in the spectrogram annotation). However, the pitch level in such tokens is lower and flatter than usual H spans, to the point that one might also transcribe the sequence as ending in a string of mid tones. Here this will be reported using the “non-falling” diacritic, viz, [na kodeeke^o]. This is not a distinct underlying tone, it is a way of capturing a fact about the phonetics of a given token. The phonetic characteristic of L*^o is that all syllables from the last H are more or less at the same pitch, which is noticeably lower than a sequence of H tones.

This can be compared to [na koté] ‘we will bury’, whose final syllable is higher than the preceding ones and pitch falls substantially from the beginning of the syllable.



In the data of this work, some tokens may be presented as e.g. [na kodeeke^o], which is phonologically equivalent to [ná kódeéké], but use of H^o indicates that the degree of final lowering is significantly less than in tokens such as [na koté].

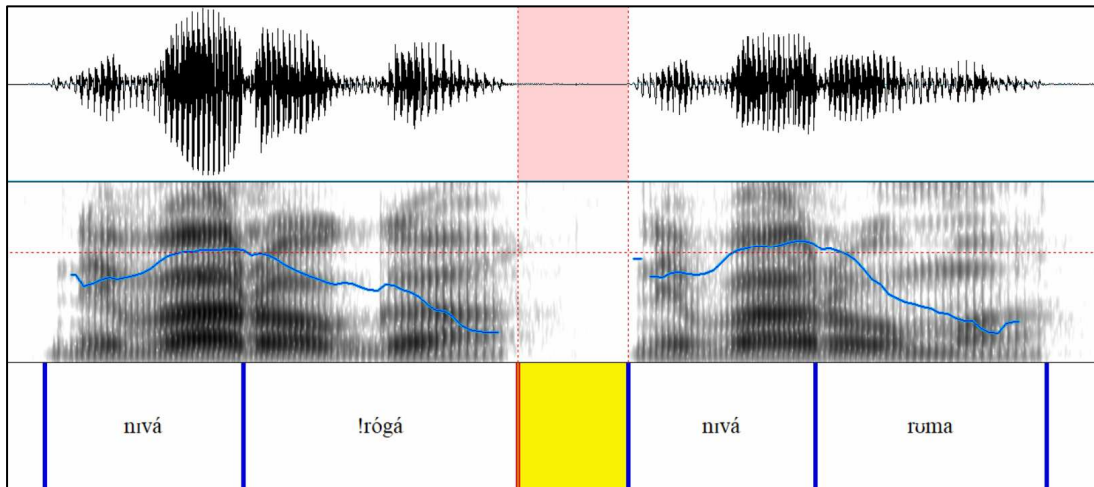
An example of the non-contrastive phonetic distinction between H¹H* and HL*^o is seen in the following pair of words [avá¹rííráná] ‘the ones who will come back’ versus [ɪpúúmba ya vayiingira^o] ‘the house which they will enter’ (note that these are separate utterances combined into a single display for convenience of reference).



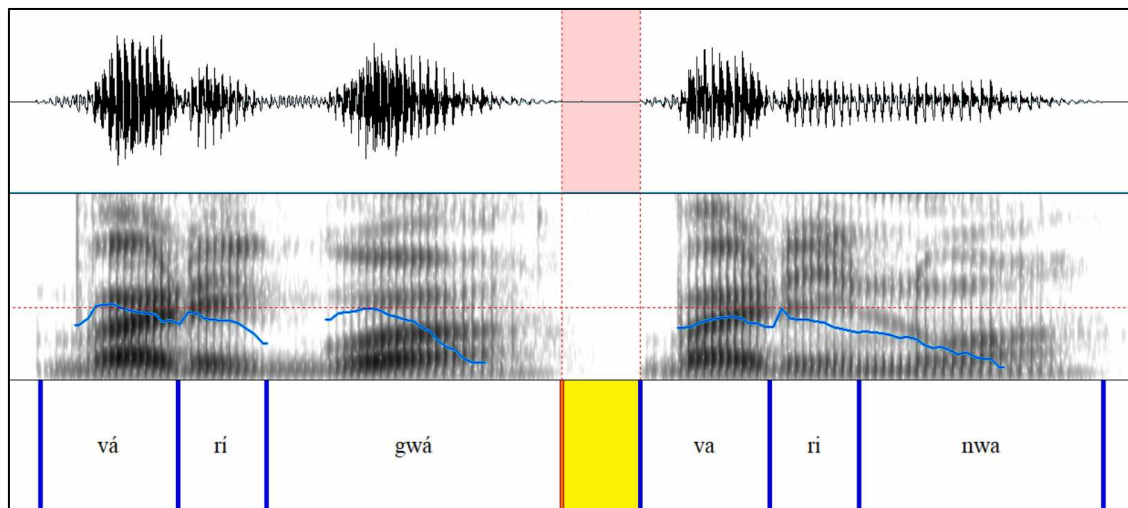
The pitch of the two is comparable and they are phonologically identical – H on the final and second syllables, and a span on toneless syllables. They differ in the pitch of the final syllables, ¹H* of the first utterance being higher than L*° of the second utterance.

So far we could say that sometimes final H has pitch level which may spread to preceding toneless syllables. However, there is also a “non-falling H”, restricted to [dá°], a variant of the negative marker [dáave], and in light of the more systematic and contrastive status of this tone variant (discussed in the next section), one might also consider “non-falling” to be an autonomous phonological property – signalled with the diacritic °.

The analogous distinction between final LL and ¹HH is also subtle. As seen in the pair [ni vá¹rógá] ‘if they bewitch’ vs. [ni várúma] ‘if they bite’, there is a sharper fall in pitch after H in HLL, but even in H¹HH there is a trend where pitch falls finally.

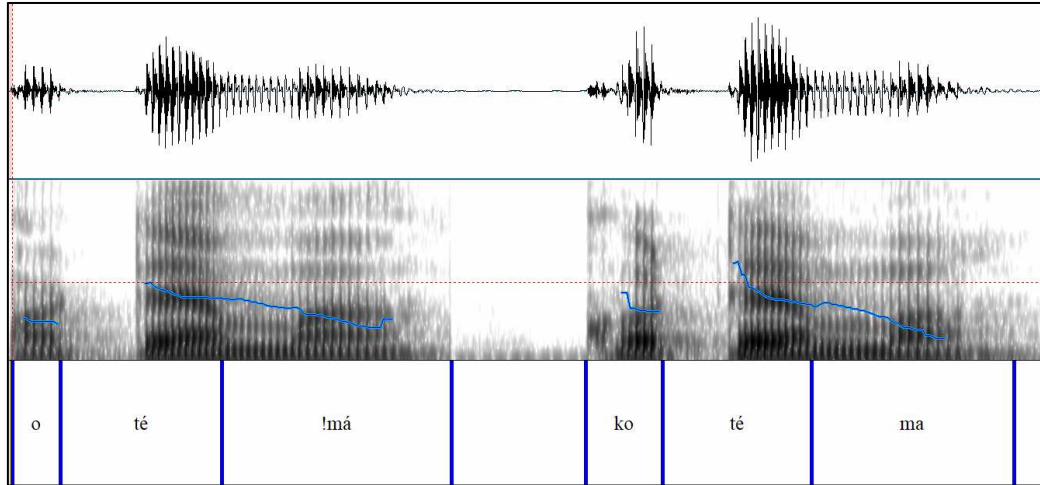


The difference between final H* and L* is even more subtle, as the following contrast between [várígwá] ‘they will fall’ and [varínwa] ‘they will drink’ shows.

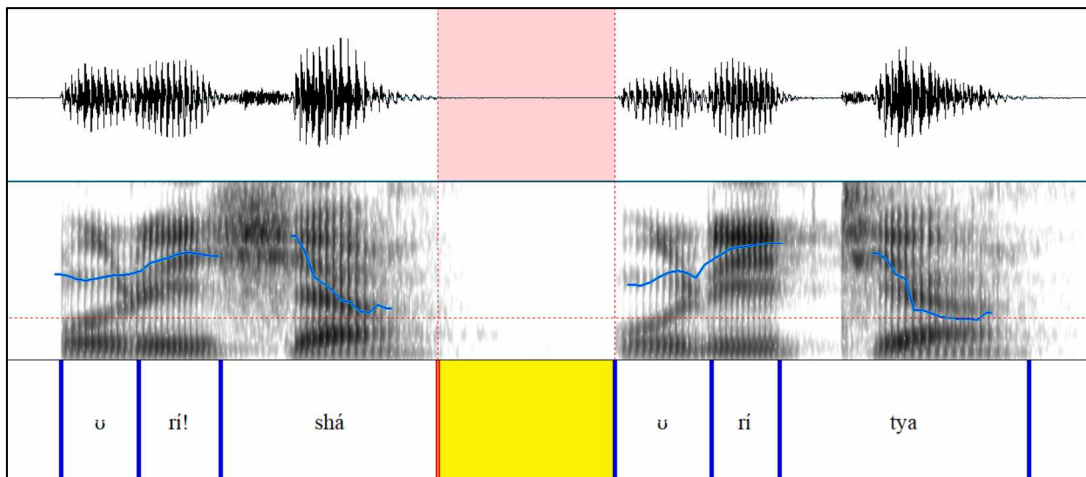


In [várígwá], the three syllables have the mean pitch 105-100-97, where [varinwa] has the pitch profile 100-100-88. The main difference lies in the extra-low pitch of prepausal L in [varinwa].

Finally, in the following case of [oté¹má] ‘he who will chop’ versus [kotéma] ‘to chop’, the phonetic difference between ¹H after H versus L after H is primarily in the greater fall in the final syllable – 120-94 in the case of [oté¹má], 115-81 for [kotéma].¹⁸



Another example of prepausal HL versus H¹H is *orí¹shá* ‘one who will grind’ vs. *orít¹ya* ‘one who will fear’. In this pair, pitch ends lower in *tya* (86Hz over the last 50 msc) and reaches that low point halfway into the vowel, compared to *shá* where pitch falls throughout the vowel to 90Hz at the final 50 msc, without a sustained level-pitch span. The perceptual effect is that *tya* sounds lower than ¹*shá* (95Hz over the syllable versus 102Hz).



¹⁸ The appearance of a greater fall in the syllable [té] in [kotéma] comes from a pitch-tracking artifact right at the consonantal release. The pitch of the utterances is the same (134Hz) at 20 msc after the consonant release.

3.2.3. PHONOLOGICAL SUMMARY OF TONAL PHONETICS

The phonological analysis of tone presented in this work depends on categorizations of pitch events in terms of a small number of tone classes. It is important to be clear on uncertainties in this classification.

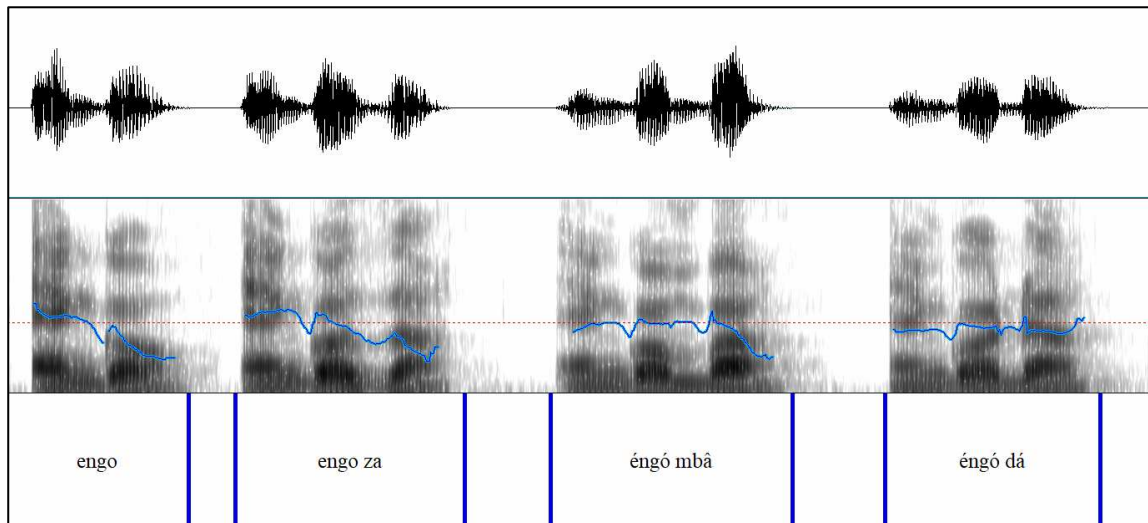
As commented on above, the answer to the question of whether there is phonological leftward spreading of H tone, as opposed to phonetic pitch interpolations, is not immediately obvious, but as we will see in subsequent chapters, the case for a phonological treatment of the facts is good enough to say in advance that there is such a rule. The main argument for a phonological treatment is that Leftward Spreading shows the indicia of surface contrastiveness – not an underlying contrast, but a derived contrast by way of different limits on the extent of spread depending on grammatical context, one that does not submit to elimination by appeal to purely surface phonetic environment. At the same time, those facts do not preclude the possibility that there is both phonological spreading and gradient phonetic interpolation. Phonetic interpolation (or similar effect) is credible, indeed necessary, for two kinds of facts. First, we have seen examples where there is a more gradual rise in pitch up to a certain syllable, often (but not always) the last syllable with H, a pattern which is attested infrequently, and which cannot be easily covered as a ‘minor detail’ variant within categorial pitch raising which is most common. Second, there is a rather frequent ‘start-up’ effect where the first potentially-H syllable of an utterance has noticeably lower pitch. It is possible that the utterance-initial syllable is optional excluded from leftward spreading; it is also possible that there is no such exclusion, but that initial HH is realized with a somewhat lower pitch on the first syllable. This constitutes a basic limit on the phonological precision of the tonal data: there may or may not be a phonologically-encoded limit on leftward spreading.

A related area of murkiness in the phonological vs. phonetic divide for tone is pre-penultimate boosting of the rightmost H in, e.g., [rwá varitagávóránya] ‘when they won’t divide up’ where [vó] has a higher pitch than [gá]. In terms of the phonological derivation, this derives from /rwá varitagavóránya/ via (stem-domain limited) phonological leftward spreading. But since [vó] has much higher pitch (supposedly because of “pitch boosting”), how do we know that [ga] isn’t simply toneless, and the higher pitch on that syllable comes from a low-level anticipatory process with a window of one syllable? Sometimes, the degree of pitch raising on [ga] is small enough that this is a plausible account, so in fact there is a significant and fundamental uncertainty in the proper treatment of apparent ...LLHHLLL..., which might also be ...LLLHLLL...

Prepausal position also carries with it certain uncertainties. Prepausal syllables usually fall to some extent (the notable exception being ...CVCV°, a variant of prepausal ...CVCV̌ where the final syllables are all at more or less the same low pitch without final fall), which gives the final syllable an overall lower perceived pitch level. Downstep between two separate Hs has that same effect, therefore one may wonder whether a prepausal H¹H-like pattern is truly H¹H, or is it HH with phonetic lowering of the second H being a biproduct of prepausal pitch fall. To decide whether final H¹H is properly analyzed (in some or all cases) as HF, we have to undertake some phonological analysis. A first step towards deciding the question is to compare phonological /LH###/ versus /HH###/. In order to do this, one has to have a basis for treating a particular word as having /LH###/ as opposed to /HH###/.

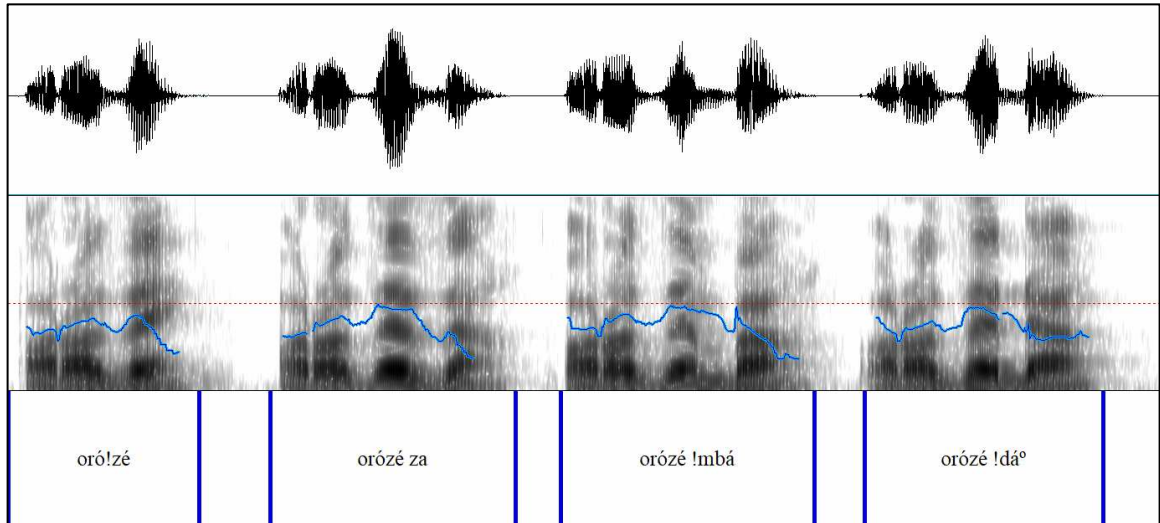
The difficulty with pursuing this reasoning is that forms with demonstrable final H are hard to generate. Monosyllabic roots can have underlying H tone, and when they do (in prepausal position), final H usually but optionally splits into H¹H, thus we have a general pattern exemplified by [íkí¹vwí ~ íkívwí] ‘fox’. Any monosyllabic noun with prepausal H potentially splits that H into a H¹H sequence: there are no lexical exceptions in the form of non-splitting final H. The same is true of lexically-H CV verb roots in the M1 tone pattern, e.g. [aránwá ~ arán¹wá] ‘he will drink’. Not every final H does so, see melodic M2 H in *varikarágá* ‘they will judge’ or M3 *vávégé* ‘that they shave’. Tokens like [íkí¹vwí] vs. [íkívwí, vávégé] provide a baseline for distinguishing final HH from H¹H, and the fact that melodic H tones behave differently from lexical H tones of monosyllabic roots suggests that this process creating H¹H is phonological.

It is generally true that a prepausal H falls to some extent, which suggests a possible account of final H-splitting as one end on a continuum of final fall *qua* phonetic process. In that view, /orozé/ undergoes Leftward Spread giving *orózé*, then prepausal H becomes falling tone, thus [orózê]. This may be perceived as non-split HH – [orózé] – if the degree of fall is small but as split H¹H – [oró¹zé] – when the fall is greater. It is true that almost all final Hs have some degree of prepausal fall, which might suggest that a phonetic process is at work. But there is one exception (and only one, as far as I have determined), namely that the truncated form of /dáave/ ‘not’ is [dá] *without* falling tone. Notice the difference between *mbá* and *dá* (both meaning ‘not’) as well as L tones *za* ‘just’ before /engo/ ‘leopard’.



As expected, there is considerable fall in the LL and LL#L sequence of the first two examples. Pitch is relatively flat in the two HH#H sequences, but there is a clear difference between the falling H of [mbâ] and the level H of [dá]. This suffices to either establish the phonological nature of final fall in prepausal H tones, or else it establishes that [dá] has some property yet to be determined, which somehow foils expected final fall. This work will adopt the latter stance (*dá* is an exception in some sense) and transcribe the truncated form of *dáave* as [dá⁰]. This transcription means that the pitch property of this

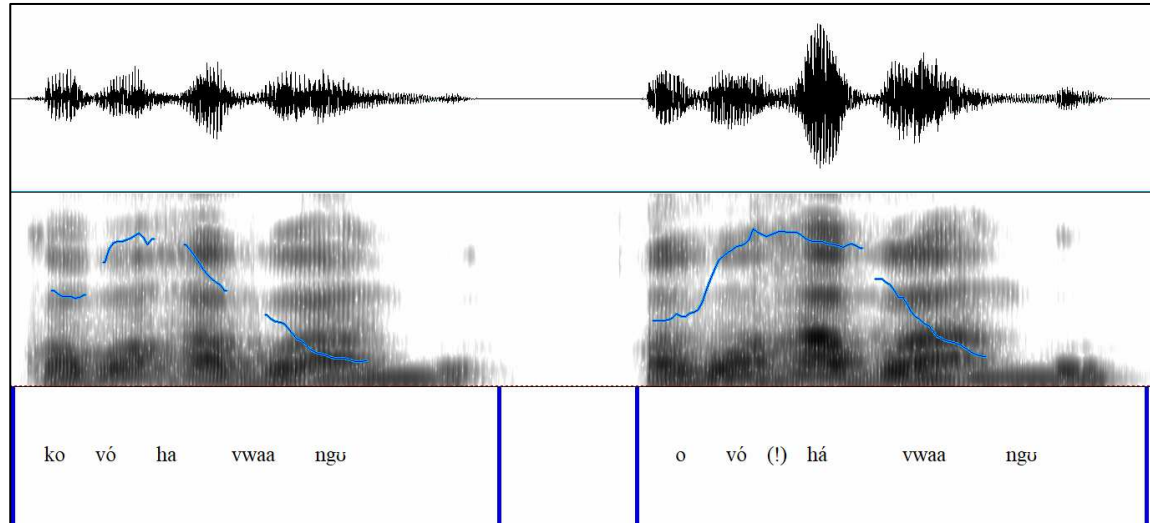
syllable is linguistically significant and must be recorded in the data, without burying the analysis of the contrast in the transcription. The examples below further exemplify the difference between the split-H citation form [oró¹zé] and the same noun before /za/, /mbá/ and /dá^o/. The difference between [orózé !mbá] and [orózé !dá^o] provides a perceptual basis for phonetically distinguishing downstep from prepausal fall.



The upshot of the contrast between [mbá] and [dá^o], and the fact that systematic H-splitting has a grammatically contrastive distribution (*vávégé* vs. *oró¹zé*), is that H¹H cannot be dismissed as a phonetic variant of HH. The downstep in *oró¹zé* is not phonologically very robust – it only occurs pre-pausally, and is optional (though most frequent). There are other instances of final H!H which are more robust.

Final H¹H is also created in verb inflections where under certain circumstances, the stem can have an initial and a final H, for example [orí¹kámé¹né] ‘the one who would live’ cf. [oríkaká¹ráángé] ‘the one who would fry’. In this verb form, the stem has a initial H and a final H, the two being separated by downstep as expected. In the case of a CVCV stem, this means that a H¹H structure will be created – the tone pattern of [orí¹kámé¹né] is phonologically analogous to that of [oríkaká¹ráángé], save for the number of syllables. Similarly, in the bare future headless relative verb form, the initial H in /ó-vohá/ shifts to the right in [ovó¹há] ‘the one who will tie’, cf. /áva-vohá/ → [avá¹vohá] ‘the ones who will tie’. This contrasts with /ari-goná/ → [arigóná] ‘he will sleep (indefinite future)’ without downstep between the two surface H-toned syllables. In examples like [arigóná], the final syllable has the perceptually highest pitch, and may sound like [arigōná] or [arigóná], but not *[arigónā] ≈ [arigó¹ná].

The H!H tone structure of [ovó¹há] is contextually robust, in that the pattern is still found phrase medially. Compare [ovó¹há vwaangu] ‘the one who will tie quickly’ and [kovóha vwaangu] ‘to tie quickly’.



We see that in [ovó¹há vwaango], the pitch level of the stem syllables [vó¹há] are similar, indeed there is not a clear drop of pitch as encountered in most ...H¹H... sequences.¹⁹ Thus we have good reason to say that [ovó¹há vwaango] has independent H tones separated by downstep, and apparent final H¹H in this context is not the product of phonetic final lowering.

While we can identify invariant H¹H as having separate Hs and distinguish them from prepausal-only H¹H as in /orozé/ → [oró¹zé], we cannot argue directly that lowering of zé proves that prepausal H splits into two Hs. But we know from examples like /engo mbá/ → [éngó mbá], where the pitch-lowering percept downstep is lacking when prepausal H does fall, that final H¹H is not a misanalysis of the falling tone of final HH.

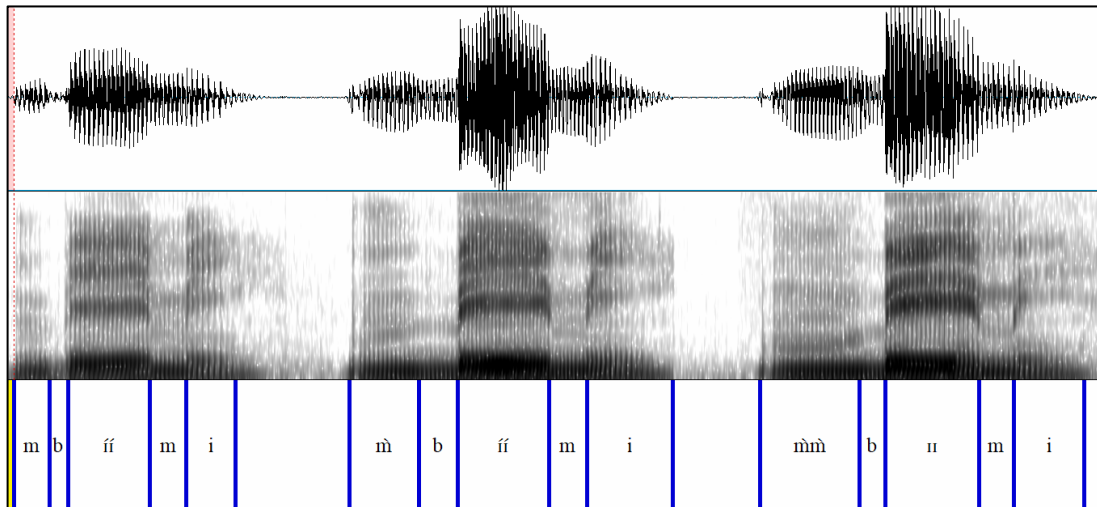
3.3. Syllabicity

While there is no underlying distinction between syllabic and non-syllabic segments, there are derived syllabic nasals as found in *mgera* ‘river’, which is phonetically [mgera]. Syllabic *m* derives from /mʊ/ by rules discussed in X, and [mgera] can also be realized as [mʊgera]. In this case, the fact of being syllabic can be easily recovered on the surface from the fact that the nasal and following consonant are not homorganic, whereas all non-syllabic nasals are homorganic with the following consonant. Reduction of prefixal /mʊ/ is particularly frequent, indeed almost universal and possible obligatory for many speakers before labials *p b v f m*. A tactic of reading syllabicity off of the homorganicity of the nasal and following consonant does not consistently work. In certain cases, a labial nasal

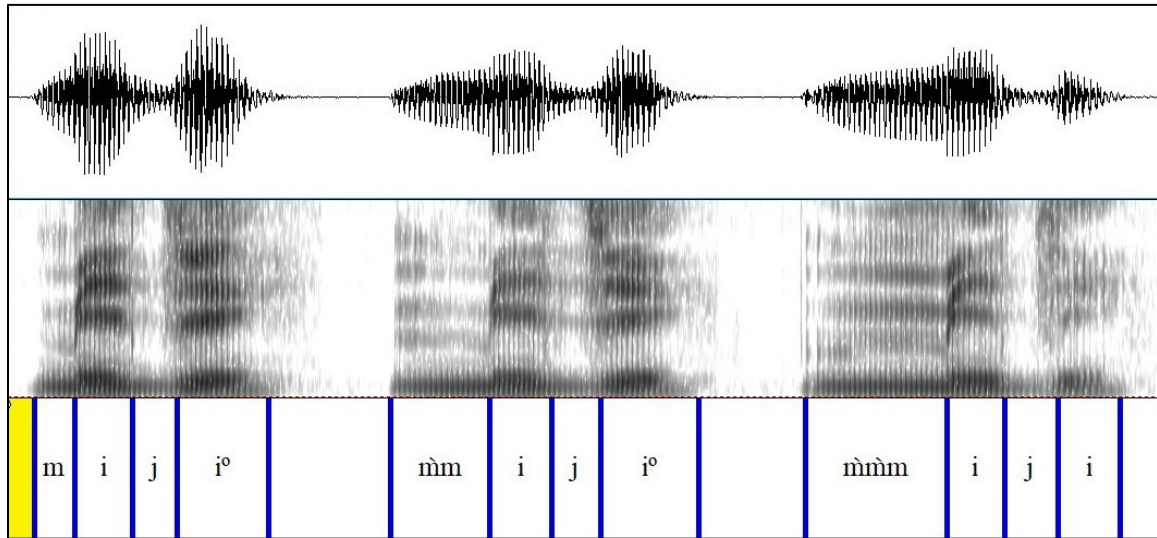
¹⁹ Most examples of ...H¹H... as observed in the preceding sections also have the effect of Leftward Spreading so that we encounter ...HHH¹H... Because the pitch peak in a multisyllabic H span is always at the right edge of the span, we can tell that [ovó¹há vwaango] is not actually [ovóhá vwaango]. If the H of [vó] were actually the result of Leftward Spreading, the pitch of [há] would be significantly higher than that of [vó]. Phrase-medial ...LH¹HL... has a very limited distribution, which is why we have not previously encountered this pattern.

plus labial always has a syllabic nasal, for example /mʊ-pííra/ ‘ball’ → [mpííra]. A nasal before another nasal, a fricative, or a voiceless stop is always syllabic. However, there is only one context where syllabicity of a labial nasal is surface distinctive, and that is before *b*, as exemplified by [mbárízi] ‘I counted’ versus [ṁbárízi] ‘2p counted’, the latter from /mʊ-várízi/. Before consonants other than [b], *m* is always syllabic, because non-syllabic *m* causes other underlying labials to change to [b] (in the case of *p* v *f*) or causes *m* to delete (in the case of *m*). Syllabicity on a (nasal) consonant is here notated with grave or acute accent, depending on whether it bears L versus H tone, thus [ṁbárízi] ‘2p. counted’, or else with an apostrophe (m'bárízi).

A syllabic nasal may be distinctively long, which gives rise to near-minimal triples such as *mbíími* ‘I measured’, *ṁbíími* ‘2p measured’ and *ṁṁbíími* ‘2p have measured’. The distinction in these nasals is realized phonetically via the greater duration of the syllabic nasal and the even greater duration of a long syllabic nasal.



Preconsonantal non-syllabic [m] has a duration of approximately 75 msc, whereas short syllabic [ṁ] has twice that duration, and long [ṁṁ] is about 1.40 times the duration of [ṁ]. A difference also exists between prevocalic non-syllabic, short syllabic and long syllabic *m*, where a root beginning with /m/ may have the 1s prefix /N/ which deletes before a nasal, vs. /mʊ/ ‘2p’, and the lengthened variant, resulting in the phonological distinction [miji°] ‘I strangled’, [ṁmiji°] ‘2p strangled’ and [ṁṁmiji] ‘2p have strangled’.



In this case, the overall duration of the nasal is longer in the case of syllabic and long-syllabic nasals where [m̩m] is about 2.5 times the duration of [m] and [m̩m̩m] is about 1.5 times the duration of [m̩m].

Syllabic *m* is reasonably common in word-final position, where it may freely vary with [m̩], for example, *ri¹mwáám̩* ~ *ri¹mwáám̩ó* ‘dark₅’. This, again, is the result of reduction of /m̩/, in this case without a following consonant. There do not seem to be any cases of mandatory final [m̩] (H or L toned), though some words like *mááram̩* ‘crushed rock for roads (murrām)’ are almost always produced without final [̩]. The word ‘needle’ is produced by BK as *isíindaani* or *síindaan̩*, and otherwise syllabic *n* does not occur finally.

Reduction of /#mC/ to [̩ ~ n̩] is uncommon except for speaker BK, but is sometimes attested with other speakers.

reduced

[bk] *ín¹gúróve*
 [bk] *ín¹dógónyi*
 [sy] *ínzókí*
 [em] *ínz-ótá¹dééchí*
 [em] *ínógv-íínáv¹ú*
 [bk] *innéke*
 [sy] *inzógu*
 [em] *ingórové isíru*
 [em] *inzí¹ndátáádééka*

unreduced

ín¹gúróve
ín¹dógónyi
ínzókí
ínz-ótá¹dééchí
ínógv-íínáv¹ú
innéke, rinéke
inzógu
ingórové isíru
inzí¹ndátáádééka

‘pig’
 ‘ant sp.’
 ‘bee’
 ‘I who didn’t cook’
 ‘sewn clothes’
 ‘Syzygium cordatum’
 ‘elephant’
 ‘stupid pig’
 ‘I who didn’t cook’

Although nearly all cases of geminate syllabic *n* derive from optional reduction of an initial syllable, the word *n̩nya* ‘mother’ is a candidate for having an underlying syllabic nasal. One token from speaker FA presents this word as *nyina*. In a very few other tokens,

the word is given with initial [i], viz. _[bk]*ínnyá móráhu* ‘good mother’, _[em]*ínyé’ évé* ‘his mother’, _[em]*inyó’ óvó* ‘your mother’, but otherwise this word is simply [ɲnya].

4. Syllable structure

Syllables in Logoori usually have an onset consonant, though syllables may begin with a vowel at the beginning of a word. The stricture in *y* before *i* is sufficiently reduced so that it is often imperceptible, e.g. *omó’yáá(y)i* ‘boy’, *rivógo(y)i* ‘amaranthus sp.’, *ligúú’yágúú(y)i* ‘dragonfly’, *dá’nibóói*. Other apparent vowel sequences include those in *lyá’óa* ‘flower’, *ring’ó’ááni* ‘crested crane’, *ibía* ‘beer’ and *omójá’lúó* ‘Luo’, which may also be interpreted as *lyá’(w)ú(w)a* ‘flower’, *ring’ó’(w)ááni* ‘crested crane’, *ibi(y)a* ‘beer’ and *omójá’lú(w)ó*. Such an interpretation is possible since underlying prevocalic *w* is nearly non-existent in the language, existing only in *kaháwa* ‘coffee’, *rowáaya* ‘wire’.²⁰ A final example of possible vowel sequences is *ikibi’ráóóni* ‘saucer’, which however has *h* after *a* for some speakers (*kibi’ráhóóni*).

Syllables maximally begin with NCG where N is a nasal homorganic with the following consonant, and G is one of the glides *w* and *y*. The nucleus of the syllable is a single vowel, which may be long or short. Other consonant clusters exist in loanwords, for example *ibóósta* ‘post office’, *kondákta* ‘conductor’, *ó’mstáári* ‘line’: it is unknown whether such clusters are modified in the speech of Logooris with no knowledge of English.²¹ Some speakers also have word-final consonants in loan words such as *ikóreet* ‘crate’, *imíishen* ‘mission’, *itáp* ‘tap’, *skól*, *iskol* ‘school’, *chíf* ‘chief’, *idháhaab* ‘gold’ (Swahili)²² though can have a final vowel as well (*isokóóru*, *imíisheni*, *chífo*). Final consonants are also occasionally attested in native words, where a vowel has been deleted, e.g. *ovwéeref(u)* ‘heaven’, *ovó’cháaf(u)* ‘filth’, *iring’ówaan(i)* ‘crested crane’, *icháany(i)* ‘house site’, *i’ndógony(i)* ‘ant sp.’ *igá’rádáas(i)* ‘paper’.

There are no generally-applicable phonological tests that establish how many surface syllables exist in *ó’mstáári* (3 or 4?), *lími* (2 or 3?), *ki’mwáám* (3 or 4?). Insofar as syllabic consonants derive from the reduction of CV syllables, it will be assumed that those processes are syllable-internal reorganizations which do not reduce the number of syllables.

The status of geminate consonants as reduced syllables is a matter of theoretical phonological analysis. The first half of a geminate can be tone-bearing, for example /kó-rí-ríma/ is often realized as [koílíma], where pitch rises over [l]. Syllables only have one or two moras, but a syllable can contain a long vowel plus a syllabic or geminate consonant – if indeed we are to assume that [vaàm] in [vaàmmijí] ‘they strangled him’ from /va-a-mu-migí/ is a single syllable. Parsed as [vaa.m.mi.jí] there is nothing worth commenting on, in terms of moraic count in any syllable.

²⁰ That is, most instances of [w] derive from morpheme-final /u/.

²¹ It is equally unknown if there exist (adult) speakers of Logoori with no knowledge of English.

²² *dh* is a voiced dental fricative, used by some speakers in loanwords from Swahili and English.